Energy Reports 12 (2024) 75–94

Contents lists available at ScienceDirect

Energy Reports

journal homepage: www.elsevier.com/locate/egyr

Research Paper

A novel strategy to enhance power management in AC/DC hybrid microgrid using virtual synchronous generator based interlinking converters integrated with energy storage system

Hayder Oleiwi Shami [a,b], Ali Basem [c], Ameer H. Al-Rubaye [d], Kiomars Sabzevari [e,*]

[a] Department of Accounting, Al-Amarah University College, Maysan, Iraq
[b] College of Administration and Economics, Department of Economics, University of Misan, Iraq
[c] Faculty of Engineering, Warith Al-Anbiyaa University, Karbala 56001, Iraq
[d] Department of Petroleum Engineering, Al-Kitab University, Altun Kupri, Iraq
[e] Department of Electrical Engineering, Technical and Vocational University(TVU), Tehran, Iran

ARTICLE INFO

Keywords:
Hybrid microgrid
Virtual synchronous generator
Interlinking converter
Energy storage system

ABSTRACT

Hybrid microgrids (HMGs) have a flexible structure, higher reliability, and both AC and DC MG merits. However, in this type of MGs, there are challenges such as accurate power sharing, the voltage control of the DC link, the AC side voltage and frequency stability, the negative impact of telecommunication delay, and so on. Therefore, a power management scheme is presented here that can perform proper and precise power sharing in a hybrid AC/DC MG. This HMG consists of one energy storage system (ESS) along with two interlinking converters (ILC) based on a virtual synchronous generator (VSG). Besides, the physical inertia of a MG is less than that of a power grid, hence, changing the output power of wind turbines and photovoltaic (PV) arrays might make system unstable. To overcome this challenge, virtual inertia is used. As the virtual inertia can be embedded with the VSG method, in this scheme, the VSG strategy is implemented to enhance the system dynamic behavior by imitating the synchronous generator kinetic inertia. The VSG-based ILC1 converter is used to establish interlinking and energy exchange between AC and DC sub-grids and share power. Also, other interlink devices including a bidirectional half-wave DC/DC converter integrated with the ILC2 are considered, which aim to control the voltage of the DC link and exchange part of the power between the AC and DC sub-grids. An ESS is placed between the bidirectional DC/DC converters and the DC link of the ILC2. Such a structure is referred to as a two stage interlinking converter with an ESS (TSILC-ESS). The power management system employed in AC/DC HMG according to the ILC controller is utilized to form the DCMG along with controlling the TSILC-ESS, providing additional and auxiliary services to the power grid. To assess the introduced scheme of both AC and DCMG to investigate VSG-based ILC1 and ILC2 in grid-connected and islanded operational mods, various resource and load profiles are assumed. The simulation study is performed in MATLAB/ Simulink environment to confirm the proposed method.

1. Introduction

A microgrid (MG) denotes a group of loads, renewable energy resources (DERs), and energy storage devices (ESDs), operating as a controllable generation unit and can work in both grid-connected and islanded modes (Parhizi et al., 2015). Characteristics such as possessing a MG unit controller and the high capacity of the MG considering the critical peak load increase the adaptability of system and power distribution capability, resulting in improvement of the energy quality, reliability, and efficiency of the supplied power (Rocabert et al., 2012).

MGs often include technologies such as photovoltaic (PV) systems as DC sources or wind turbines as AC sources, which need DC/AC/DC or DC/AC interfaced converters to connect the grid. These inverters can play important role in voltage and frequency control of islanded MGs and facilitate MGs participation in outage schedules. For islanding and synchronization issues, the static disconnect switch (SDS) would be the main component of the MG which swiftly disconnects the MG from the power grid in the circumstances such as overvoltage, undervoltage,

* Corresponding author.
E-mail address: ksabzevari@tvu.ac.ir (K. Sabzevari).

https://doi.org/10.1016/j.egyr.2024.06.001
Received 24 March 2024; Received in revised form 11 May 2024; Accepted 1 June 2024
Available online 12 June 2024

high/low frequency and overcurrent (Unamuno and Barrena, 2015a; Nejabatkhah and Yun, 2014; Hirsch et al., 2018). Considering the loads and resources connected to MGs, they are classified as AC, DC, and hybrid AC/DC MGs. Considering the stability requirements in the electrical grids, AC microgrids (ACMGs) can be considered as a suitable option for managing the energy supplied in industrial plants (Sarangi et al., 2020). Also, the stability issue is associated with improving the performance of converters,thererfore, they are categorized as grid-forming, grid-feeding, and grid-supporting units (Rocabert et al., 2012; Engler, 2004).

The evolution of power electronics enables the development of DC equipment such as ESDs, DC loads, and DC resources that work with each other, forming a DCMG. Compared to ACMGs, the merrits of DCMGs include improved efficiency, absence of reactive power, and their simple control methodologies (Sarangi et al., 2020; Nordman and Christensen, 2016). The different architectures and structures of DCMGs are presented in (Sarangi et al., 2020; Elsayed et al., 2015). Advancements in microgrid technology show that the integration of RERs, coupled with the implementation of smart grids infrastructure (using smart sensors and energy management systems), holds the potential to provide energy systems with higher reliability and efficiency more cost-effectively. Enhancing the efficiency and reliability in electrical grids is gained through the implementation of DC distribution within MGs. DCMGs represent an appealing technological solution in contemporary power systems, primarily owing to their seamless integration with RERs, loads, and ESDs. Recently, research about DCMGs has increased, which approaches them more to real-world implementation. Reference (Elsayed et al., 2015) represents the latest DCMG technology and covers aspects such as communication systems, power quality concerns, potential grounding strategies, architectural considerations, and AC interfaces. The DCMGs benefits can be found across a wide range of implementations to enhance the respective efficiency and reliability.

One of the main goals in both AC and DC microgrids is power management. Nowadays, various methods and strategies have been proposed to manage the power of islanded and grid-connected microgrids. In (Wen et al., 2018), different strategies of power management in islanded ACMGs are investigated. This reference introduces a control scheme for energy management in autonomous ACMGs that incorporates a hybrid energy storage system, comprising both supercapacitors (SC) and batteries. With respect to the SOC of the battery, the PV unit operate in the load power tracking mode or the maximum power point tracking (MPPT) mode, so that avoids the battery overcharging.

In (Neto et al., 2020; dos Santos Neto et al., 2020), power management strategies are proposed by employing virtual inertia for DCMGs. To manage power flow of a grid-connected DCMG, a power management strategy is presented in (Neto et al., 2020). The model of the microgrid consists of an energy storage system (ESS) including a battery bank and a bidirectional dc-dc converter that acts as a grid-supporting converter. In addition, this microgrid includes a VSI based ac power system that acts as a grid-forming unit, a DG that acts as a load regulation unit for the consumers. The power management method utilizes the concept of virtual inertia, which is according to the management performance based on charge to adjust the battery bank charge and discharge process with respect to the power flow in DCMG's. Therefore, flowing the high powers is avoided and as a result, the lifelong of ESS batteries increases. Through the proposed power management, an independent load flow can be attained in the DCMG, however, the AC power system is entirely focused on the DC grid-forming as well as managing the excess/deficit of power. Moreover, the presented method can simplify the communications among the ESS and the grid inverter, as the power of VSI is only based on the exchanged data (dos Santos Neto et al., 2020). Conventional power management methods in microgrids are based on two important control methods including droop and Virtual Synchronous Generator (VSG). Impressive studies have been conducted regarding droop and VSG controllers in the literature. As DGs consisting of PV systems do not involve any rotating parts to generate inertia, therefore, they can contribute to supporting the frequency of the grid through increasing virtual inertia by using electronic inverters, while in conventional power generation units, the synchronous generator (SG) provides frequency support during disturbances. However, compared to conventional power generation units, different characteristics can be observed because of the rotating mass of those DGs that are electronically connected to the power system or grid. The primary side generated power of electronic interface based DGs can be adjusted by inverters, however, they cannot provide the needed inertia and damping for the system. As a result, electronic interface based DGs cannot improve the system stability. However, this problem is solved through implementing appropriate control techniques to the grid-connected inverters and controlling their switching in such a way that it works as a virtual synchronous generator by imitating its behavior. In the realm of grid-connected inverters, there exists a category known as VSGs that imitate the steady-state and transient attributes of a synchronous generator. Based on predictions, the future power grids will include integrated VSG systems. Consequently, it would be crucial to analyze and enhance the VSG-based grid transient stability.

The fundamental concept of VSG as well as various methodologies is elucidated in (Zong et al., 2023a), which can connect the grid to the inverter so that imitates the operational characteristics of the synchronous generators. In addition, a comparison between the performance features of synchronous generator and common VSG is made, like Q-δ, and P-f. As a result, the introduced VSG strategy imitates the swing relationship related to the conventional synchronous generator, while the technique presented in another study reflects the appropriate operational performance of the synchronous generator. Nevertheless, the VSG controlled by current can be considered as a current source. Therefore, supplying frequency and voltage support to the system presents a tremendous challenge. Consequently, a technique as VSG controlled by voltage is offered to overcome the drawbacks of current-controlled VSG. Simulating the inertia of rotor and modulation features of the frequency of the synchronous generator system in frequency controllers is considered as the inherent nature of voltage-controlled VSG that can obtain frequency stability improvements. At the same time, the voltage-reactive power equation is mainly assumed in the voltage controller, so that controls the reliable voltage output. The power controller and voltage/frequency controller cause VSG to perform power control and frequency modulation functions. In addition, several other methods develop the model and controller to simulate various dynamic controllers of synchronous generators. However, VSG control has not yet developed enough in theoretical and practical applications compared to a conventional synchronous generator (Silveira et al., 2021).

On the contrary, many methods based on droop control to manage MGs have been reported in the recent works. Ref (De Matos et al., 2014). proposes a droop based controller to control the power generated by RER in an islanded MG. Here, a power grid comprises a battery bank powered a VSI as the grid-forming converter, a VSI based wind turbine as grid-feeding, and loads. The chief goal in this study is controlling the battery bank's state of charge (SOC), which limits the terminal voltage of battery by control of the RER generated power. Hierarchical control strategies are used in ACMGs (Rocabert et al., 2012; Li et al., 2019). The operation of the paralleled VSIs within an ACMG are controlled by a hierarchical controller based on distributed concept (Li et al., 2019). To control the VSI output voltage, the common two-loop proportional-integral (PI) controller shows poor function during the parameter variation, and a narrow band of stable performance is observed. Therefore, the inner loop current controller is based on sliding mode control (SMC) (zero level) and outer loop voltage controller is based on combined H_2/H_∞ (level 1), controlling the VSI output voltage, benefiting from the robust and quick transient response features. The droop controller (level 2) is employed to prevent circulating current between VSIs without using any extra communication links. An effective way to address frequency and voltage variations induced by the droop control strategy while concurrently optimizing active power sharing is proposed in (Li et al., 2019). This is achieved through a multi-agent system (MAS)-based approach for distributed automatic voltage control (AVC) and distributed economic automatic generation control (AGC) at level 3.

Distributed hierarchical control is discussed in (Hou et al., 2018) for grid-connected, islanded, and transition modes. A coordinated multi-layer control strategy is presented in (Mahmud et al., 2019) to manage grid-connected ACMG energy. This strategy can predict the load demand of customers and generated solar power for managing the energy in the next day. This method utilizes solar power and bidirectional power exchange between electric vehicles and batteries to support load by a combined response. Moreover, this scheme anticipates any uncertainty in customer demand and power generation and employs preventive actions to cope with this uncertainty. In (Li et al., 2017), droop based decentralized controllers are investigated in DCMG without telecommunications among converters. Furthermore, a dc voltage droop controller based on observer and feed-forward current controller is presented for a DCMG, which can effectively enhance the dc voltage control dynamic response using the observer. In addition, the observer can provide both feed-forward and feedback current for the current and dc voltage droop controllers without extra current measurements. Moreover, in comparison with the conventional droop schemes, the capability and stability are improved.

Multi-layer strategies based on hierarchical control are presented for DCMG in (Han et al., 2019; Jin et al., 2013). The coordinated controller for multiple dc resources and energy storage is experimentally performed in (Jin et al., 2013) by a new hierarchical controller. The DC bus voltage basically operates as a balance index of demand and supply; ensuring a valid performance is obtained by using a wireless controller. The control strategy in the cooperative manner is investigated in (Chen et al., 2017) for power sharing. Here, a distributed cooperative control frame is developed for a DCMG, and the system small-signal stability analysis is presented. The first layer consists of a droop controller that coordinates the operation of the microgrid; the second layer operates by using a consensus formulation for regulating the reference voltage of the DC bus. By a proper design, the cooperative controller is able to reach the average consensus of the DC bus voltage. The DCMG operation during fault situations is addressed in (Shaban et al., 2023) with a focus on diverse control strategies applied to hybrid energy storage systems (HESS) in DCMGs. Using both a reference battery current compensation (RBCC controller) and an actual battery current compensation (ABCC) controller, the DCMG fault response is assessed (Rahgozar et al., 2022; Rashid et al., 2023). In addition, the straightforward equivalent circuit is investigated to balance the voltage in DCMGs throughout pole-to-ground faults.

In order to obtain the advantages of AC and DC technologies, recent literature addresses hybrid AC/DC MGs as the future structure of smart grids (Planas et al., 2015). The configuration of hybrid MGs is covered in (Unamuno and Barrena, 2015a; Nejabatkhah and Yun, 2014). The droop methods are used in (Eisapour-Moarref et al., 2019) for power sharing in hybrid MGs, while in (Wang et al., 2017), multilayer controls based on hierarchical strategy are implemented for the power converters. Similarly, VSG based strategies are presented in (Liu et al., 2019; He et al., 2019). In (Eajal et al., 2016), different aspects of control methods are categorized by an integrated control approach. Various control schemes and power management systems are presented for AC/DC hybrid MGs in more detail in (Unamuno and Barrena, 2015b; Eghtedarpour and Farjah, 2014; Gupta et al., 2017).

Hybrid MGs employ interlinking converters (ILC) that enable power sharing between DC and AC MGs (Loh et al., 2012). Assuming this purpose, such converters act in grid-forming and grid-supporting modes (Rocabert et al., 2012; Ullah et al., 2023). Ref (Tricarico et al., 2020). presents a novel ILC structure by implementing the bidirectional interfaced converters in a hybrid MG that works in islanded and grid-connected modes. A multilevel ILC converter structure is adopted in (Sadeghi and Khederzadeh, 2019) to improve voltage control by adjusting voltage deviations, in addition to mitigating the size of inductive filters. Nevertheless, the multi-level structure needs additional switches, which increases the power loss as well as complexity of control. In (Phan and Lee, 2018), an ILC converter is utilized for improving the power quality in an islanded hybrid MG with non-linear loads. One control strategy is introduced with no communications, reducing PCC harmonics and providing the possibility of power sharing among subgrids. However, the effect of voltage variation on the DC-link does not assessed. Likewise, two harmonics control methodologies are presented in (Tian et al., 2017) for ILC to reduce the distortion. These strategies are presented by virtual impedance theories, having low switching frequency, and reducing delay impacts, while affecting the resolution of the waveform. One detailed overview is provided in (Ordono et al., 2019) for control structures and techniques of ILC converter. The small-signal stability of an Hybrid MG is significantly influenced by the direction and magnitude of power passes in an ILC converter. A non-droop tracker is suggested in (Awad et al., 2023) based on an invariant-set design which is used to track the favorable time-varied reference, aiming to manage load flow within the Hybrid MG during rectification and inversion conditions. This tracker offers a rapid response, zero error for steady state, as well as resilience in case of grid uncertainties.

Various operating conditions and hybrid MG purposes have led to the presentation of different power management systems for the ILC converter (Nejabatkhah and Yun, 2014). As described in (Eghtedarpour and Farjah, 2014), a two-layer droop control is applied as a power management system to an ILC converter in an islanded MG. This modified droop controller is utilized for decentralized power sharing so that avoids the requirement of communication links among MGs or DGs. In (Navarro-Rodriguez et al., 2018), an adaptive power sharing strategy is implemented for a hybrid MG including two AC nanogrids and a central BESS. The purpose of the power management system is minimization of the need for the distribution system and extra BESS usage, also simultaneously increasing in the stability of the system in islanded mode. Nonetheless, the DC sub-grid with associated sources and loads is assumed in this study. A hybrid MG structure consisting of several AC and DC subgrids is surveyed in (Xia et al., 2017). One two-layer power management system is introduced using a sub-grid including ESS, which utilizes droop controller in the primary level for the common DC-link of the hybrid MG, and the coordinated controller is employed in the upper level. Even though droop controller is implemented, the method requires to communicate with its upper level. The authors also provide a review of the literature on power management systems focusing on power quality in hybrid MGs in (Akbari and Zare Ghaleh Seyyedi, 2023). A multi-objective optimization model is recommended in (Premadasa et al., 2023) to size and operate a hybrid energy system including various DGs, such as PV, wind turbine, diesel generator, and BESS. This paper introduces a novel concept for loading percentage optimization of the diesel generator. Furthermore, the proposed battery management index enables adaptive hour performance for the diesel generator, in addition to adjusting the battery SOC. The presented model is designed to optimized the overall costs of carbon emissions, maintenance, and investment.

In recent years, the conventional linear controllers such as PI have been replaced by nonlinear and intelligent controllers that are robust against system uncertainties. For this purpose, a review for managing energy along with Hybrid MG controllers is provided in (Ullah et al., 2023). This overview includes the employment of popular control techniques such as proportional integral derivative controllers, sliding mode controllers, fuzzy logic, and artificial neural networks. Additionally, the paper addresses maximizing of the PV power through MPPT, using an expanded artificial neural controller. This approach is compared with other control methods. Also, a novel adaptive droop control scheme is represented for multiple ILCs within the Hybrid MG (Zong et al., 2023b). This control approach not only decreases redundant power transferred among subgrids but also mitigates the droop

Table 1
Comparison of different aspects of the proposed method with some research literature.

Refs	ILCs No.	Control of ILCs			Energy storage	Power sharing and management	Reactive power compensation	Grid-connected mode	Islanded mode
		Level1	Level2	Level3					
(Silveira et al., 2021)	2	PI	Droop	PI	Yes	Yes	Yes	Yes	No
(Li et al., 2019)	2	SMC	Droop	EAGC	No	Yes	No	Yes	No
(Hou et al., 2018)	2	Droop	consensus	supervisory	No	Yes	Yes	Yes	Yes
(Eisapour-Moarref et al., 2019)	3	PI	3D-Droop	PI	No	Yes	Yes	No	Yes
(Wang et al., 2017)	1	Droop	PI	PI	No	Yes	No	Yes	No
(Liu et al., 2019)	1	VSM	VSM	VSM	No	Yes	No	Yes	Yes
(He et al., 2019)	1	VSG	VSG	VSG	Yes	Yes	No	Yes	No
Prop	2	VSG	VSG	VSG	Yes	Yes	Yes	Yes	Yes

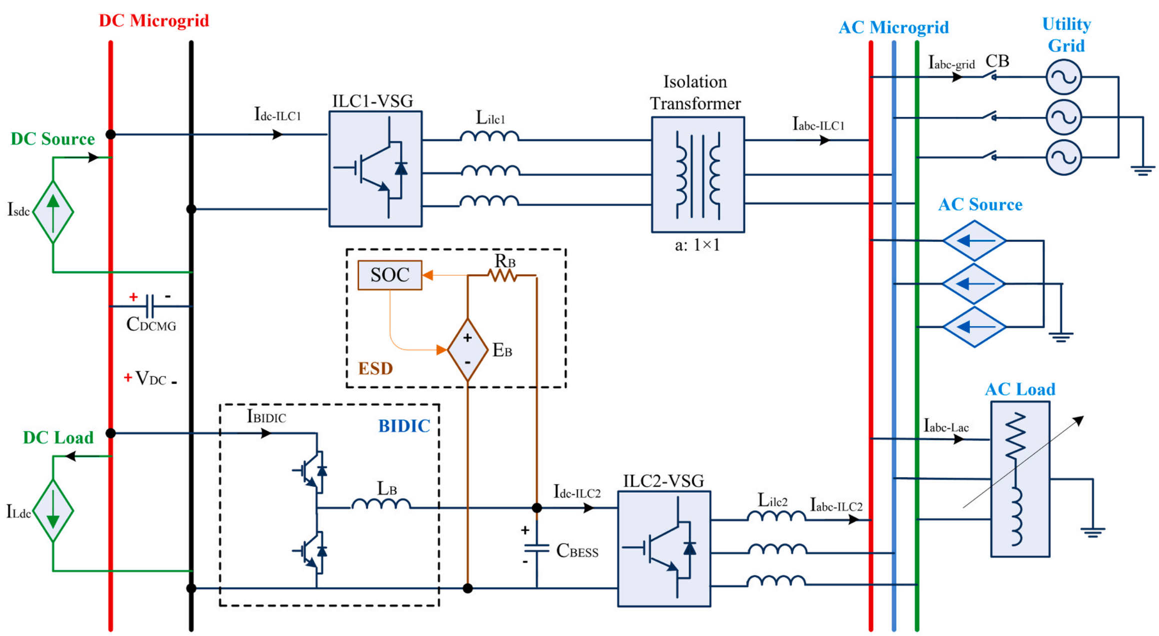

Fig. 1. Proposed AC/DC hybrid microgrid scheme.

related problems in voltages of DC and AC bus. By implementing this control method, the imprecise power sharing issues and the circulation induced by the impedance of the line or differences in capacities among ILCs are resolved (Awad et al., 2023). Furthermore, a new adaptive normalized droop control methodology is introduced in (Ullah et al., 2023) for ILCs within the Hybrid MG. While bidirectional power coordination among subgrids can be attained by the basic normalized droop technique, it suffers from low precision in transmitted power as well as subgrid bus voltage deviations from their nominal values. The presented control method mitigates redundant power flow, thereby achieving precise power sharing and effective compensation for voltage deviations. Ref (Mohamed et al., 2022) introduces an innovative adaptive control approach aimed at improving the dynamic capabilities of the ILC within a remote hybrid microgrid, all without the need for communication. The suggested autonomous method relies on the Continuous Mixed P-Norm (CMPN) algorithm to effectively regulate both the AC frequency and DC voltage of the ILC, ensuring they stay within acceptable thresholds across diverse scenarios.

As mentioned in the research literature, research gaps mentioned in the below are observed in the field of hybrid AC/DC MGs:

- The utilization of an ILC converter integrated with ESD, though a few earlier works have been conducted about this topic, but it can still work to be done.
- The power management strategy in DC and AC microgrids with centralized ESD is developing and it needs more improvements.
- The performance of a hybrid MG involving two ILCs converter with two VSG controllers in islanded mode is not studied.

According to the studied literature, the current article introduces a power management scheme for an improved structure of the ILC converters based on VSG control method. The purpose is modifying the power sharing of hybrid AC/DC MGs. This article is focused on islanded and grid-connected modes. The power factor and voltage deviation are considered as power quality parameters to be analyzed. The power factor is improved by implementing a reactive power sharing compensation methodology. A hierarchical control based on VSG is adopted that can attains power sharing and regulate the DC link voltage. Although one ILC converter can be used to achieve the above objectives, due to the presence of the power management storage (ESD) and master-slave control, two ILCs converters should definitely be used. This issue can even make the system flexible. Besides, power flow control from two separate paths increase reliability. On the other hand, DC-link voltage remains constant in all power exchange conditions by means of our proposed VSG method, while in many works, DC-link voltage needs to be dropped or stepped up for exchanging power. The assessment of the introduced power management system and the performance of the improved ILC converter structure are conducted by simulations in MATLAB/SIMULINK software. Main contributions and objectives of the paper are summarized below:

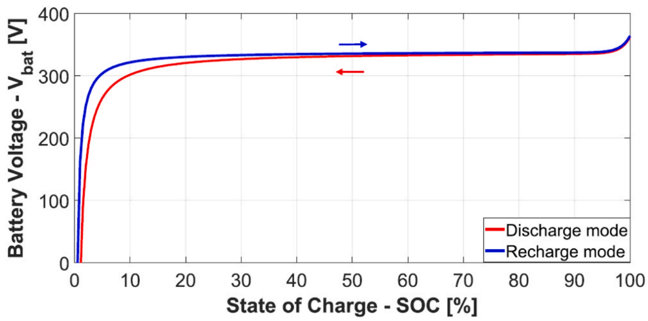

Fig. 2. Battery voltage response in terms of SOC during charge and discharge (Silveira et al., 2021).

- Using two ILCs converter with VSG controllers for each as master-slave control to improve reliability, DC-link voltage regulation and reactive power compensation.
- Using ESD to supply total loads of MG in islanded mode and when generated power from RER is reduced.
- Presenting a novel hierarchical control for power management and power sharing among ACMG and DCMG

A supplementary comparison between the method proposed in this article and the methods adopted in the research literature is provided according to Table 1.

Some merits of the proposed configuration including two ILCs are:

- Providing ancillary services such as compensation of harmonics and unbalanced voltages by ILC2 converter, while ILC1 takes over other tasks like power sharing.
- Improving reliability of microgrid by using two different paths to power flowing between DCMG and ACMG
- No need for dropping or stepping up the DCMG voltage from its nominal value to exchange active power among ACMG and DCMG, as exists in droop method.

The next sections are organized as: In the second part, the modeling and configuration of the basic microgrid along with its main components are presented. The proposed hybrid microgrid scheme using the virtual synchronous generator interface converters, the associated equations and relationships, and the controllers of the corresponding converters are presented in the Section 3. In the Section 4, the implementation of a islanded and grid- connected hybrid microgrid is simulated in the MATLAB/Simulink software; various simulations are performed to confirm the proposed strategy. Eventually, Section 5 concludes the paper.

1.1. Configuration of hybrid AC/DC MG

In this section, the modeling and configuration of a hybrid AC/DC MG in islanded and grid-connected modes is discussed. This MG consists of two ILC converters. The ILC1 shares the real power among two AC and DC sub-grids with the unit power factor (without exchanging reactive power) and operates based on the VSG controller. The ILC2 is also considered for the controlling of its DC link voltage and stabilizing of the DC voltage in the DC sub-grid with the VSG controller through a bidirectional DC/DC converter with an ESD storage; it also shares the requested real and reactive power of DC and AC subgrids' loads. In the following, the configuration and modeling of the proposed method is discussed.

The studied hybrid AC/DC MG structure of this article includes DG, electrical loads, an ESD, and the main power system. The overall structure considering the aforementioned components is depicted in Fig. 1. Each component will be explained in detail in the next subsections.

1.2. Distributed generations

The considered DG consists of renewable sources like wind and PV systems which operating based on the characteristics of wind speed and solar radiation. The frequency and voltage requirements of the power system cannot be met by the electric current produced by renewable sources. Therefore, power electronic converters are utilized to connect DGs to the microgrid. Current sources are considered to represent these converters. The powers generated by the PV and wind systems are considered as DC and AC sources in the understudied hybrid microgrid, respectively, and they are explained below.

1) the AC resources: Wind systems are a sample of AC renewable resources. Generally, they include wind turbines and back-to-back converters, allowing to inject the produced power into the power system. A regulated three-phase AC power source is considered instead of the wind system.
2) The DC resources: This resource is a PV module set linked to a DC-DC converter that operates using MPPT. For simplification of the analysis, an ideal current source is assumed instead of this source as injects power into the DCMG.

1.3. Loads

1) The AC loads: These loads are components that consume AC current, such as induction motors (Silveira et al., 2021). A balanced three-phase set of impedances consisting of series connected resistors and inductors are assumed as an AC load. Therefore, the real and reactive powers of the AC microgrid are consumed by AC loads. These loads' input and output are defined based on the desire power profiles.
2) The DC loads: Components consuming DC current, i.e. electric vehicle, DC machine, and other electronic components, are called DC loads (Silveira et al., 2021). A controlled DC resource represents the DC load, likewise the DC resource, however, it absorbs the power from the system.

1.4. Energy storage devices

Through the various available energy storage advancements, lithium-ion battery is selected as the ESD here. Shepherd's model is considered for this ESD dynamic behavior (Silveira et al., 2021); the model illustrates a controlled DC source in series with internal resistance for the battery, as demonstrated in Fig. 1. The battery bank output voltage can be simply described according to (1).

$$V_{bat} = E_{bat} - R_{bat} I_{bat} \tag{1}$$

where V_{bat} and E_{bat} consecutively represent the battery output and input voltages. R_{bat} shows the battery internal resistance; I_{bat} represents the current of the battery.

The SOC of the battery, the current, and the input voltage, E_{bat}, are correlated straightaway. It would be required to have the ratio of the battery energy consumption ($J_b = \int I_{bat} dt$) to its energy capacity (W_b) for determining the SOC, so we have:

$$SOC = SOC_0 - \frac{J_b}{W_b} \tag{2}$$

where SOC_0 is the battery initial SOC.

E_{bat} shows the main part of the battery dynamics, in which the behavior of the battery in the state of charging or discharging is expressed. The discharge model of the battery can written as (3) and (4) which expresses that of charging performance (Silveira et al., 2021).

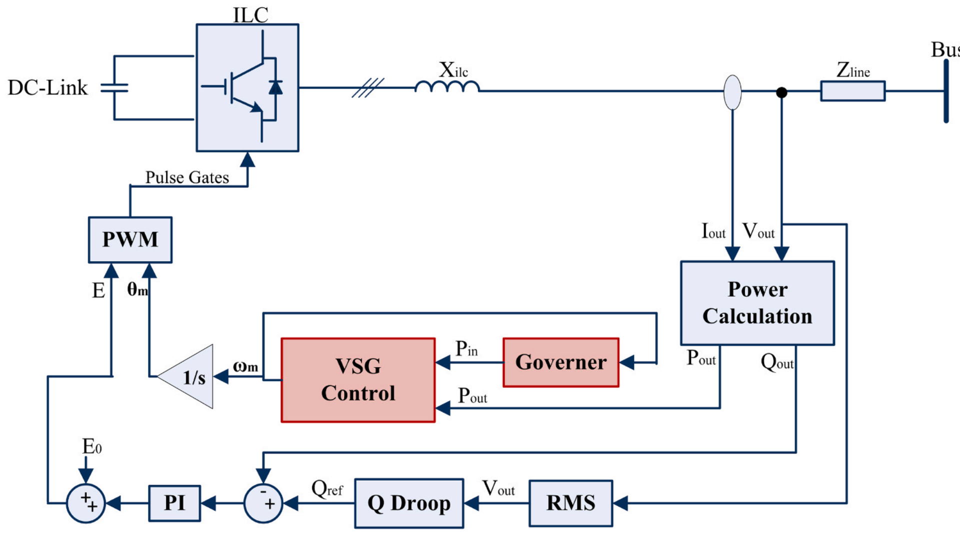

Fig. 3. VSG based ILCs control diagram block.

$$E_{bat} = E_{0b} - K_b\left(\frac{W_b}{W_b - J_b}\right)J_b - K_b\left(\frac{W_b}{W_b - J_b}\right)I_{bat} + A_b e^{-B_b J_b} \tag{3}$$

$$E_{bat} = E_{0b} - K_b\left(\frac{W_b}{W_b - J_b}\right)J_b - K_b\left(\frac{W_b}{J_b + 0.1W_b}\right)I_{bat} + A_b e^{-B_b J_b} \tag{4}$$

where K_b shows the polarization constant; E_{0b} represents the constant voltage of the battery; A_b represents the amplitude of the exponential region; B_b shows the time constant of the exponential region. The noted parameters for the Li-ion battery can be found in (Silveira et al., 2021). The output voltage of ESD, i.e. V_{bat}, is given in Fig. 2 according to the charging and discharging performance related to SOC. Notably, restrictions must be applied to the SOC for the proper operation of the battery bank. Therefore, the SOC is generally limited to 50–90 % of the overall battery charge. On the other hand, the battery is damaged if it operates outside of the SOC range, resulting in reducing the lifelong.

1.5. Power converters

In the understudied hybrid microgrid, grid-forming converters are employed in DGs that operate as current controlled sources in MPPT mode. The grid-forming converter is an ILC1 which is considered as a DCMG with a voltage controller. For power sharing improvement, two grid-supporting converters are used. These converters work as two-stage cascaded ILC (TSILC). The current work emphasizes on proposing a power management strategy for improved hybrid microgrid interfaced converters. Thus, the TSILC and ILC details are described below.

1) ILC1: The grid-forming converter of ILC1 shown in Fig. 1 (ILC1-VSG) is a VSC with an inductive filter linked to the ACMG. Controlling the DC-link voltage can be done by another converter, allowing the DC load and source connections by forming a DCMG. Therefore, ILC1 shares power among AC and DC subgrids. The dq frame is considered to describe the ILC model, as below differential relationships:

$$\begin{cases} L_i\dfrac{di_{id}}{dt} = v_{id} - R_i i_{id} + L_i\omega_i i_{iq} - v_{sd} \\ L_i\dfrac{di_{iq}}{dt} = v_{iq} - R_i i_{iq} + L_i\omega_i i_{id} - v_{sq} \end{cases} \tag{5}$$

$$\frac{C_{DCMG}}{2}\frac{dV_{DC}^2}{dt} = \left(P_{ilc1_{in}} - P_{losses} - P_{ilc1_{out}}\right) \tag{6}$$

where i_{id} and i_{iq} are the dq axis currents of the ILC1; v_{id} and v_{iq} are the dq axis voltages; L_i and R_i are the filter inductance and resistance; ω_i shows the angular frequency of ILC1; v_{sd} and v_{sq} are the system dq voltages; C_{DCMG} represents the DC-link capacity; V_{DC} exhibits the DC-link voltage; $P_{ilc1_{in}}$ and $P_{ilc1_{out}}$ are the input and output power of the ILC1, and P_{losses} is the converter losses.

By applying the Laplacian transformation to (5) and (6), and assuming the state variables, they are written as:

$$G_{c_{ilc1}}(s) = \frac{I_{id}}{m_{id}} = \frac{I_{iq}}{m_{iq}} = \frac{1}{L_i s + R_i} \tag{7}$$

$$G_{v_{ilc1}}(s) = \frac{v_{dc}^2}{P_{ref_{ilc}}} = \frac{2}{C_{DCMG}s} \tag{8}$$

where m_{id} and m_{iq} represent the modulation index for ILC1 in dq frame, respectively; and $P_{ref_{ilc}}$ is the reference power of ILC1.

2) TSILC: This converter consists of two series converters including a bidirectional DC/DC converter (BIDIC) and an auxiliary ILC2; the respective details are presented in Fig. 1. An ESD is placed between the two converters. Therefore, the BIDIC is the interface between the DCMG and ESD, while ILC2 connects the ACMG to this ESD. Since ILC2 and ILC1 are voltage source converters with inductive filters, controlling in the dq coordinates, their model is similar.

Like ILC1, the inductive current and DC link are used to model voltage the BIDIC. The direction of current is defined only if the bidirectional converter operates in the boost or buck modes. The current and voltage model of the BIDIC can be expressed the same as the ILC1 model by mathematically manipulating the equations and providing the required situations, which is described in (Tian et al., 2017). It should be remembered that the BIDIC is the DC link of the ILC1 and provides the common DC bus.

1.6. Power grid

Since the microgrid basic characteristic is its operation capability in islanded or grid-connected modes, therefore, this research exclusively studies both modes. Here, the rms voltage and frequency of the balanced three-phase AC source in the implemented power system are 220 V and 60 Hz, respectively. Each voltage source has a phase difference of 120 degrees in positive sequence compared to the adjacent phases. RL impedances are used to determine the dynamic impact of conductors and to simulate the impedances of transmission lines. These impedance values for electrical distribution systems can be obtained by the traditional short circuit power.

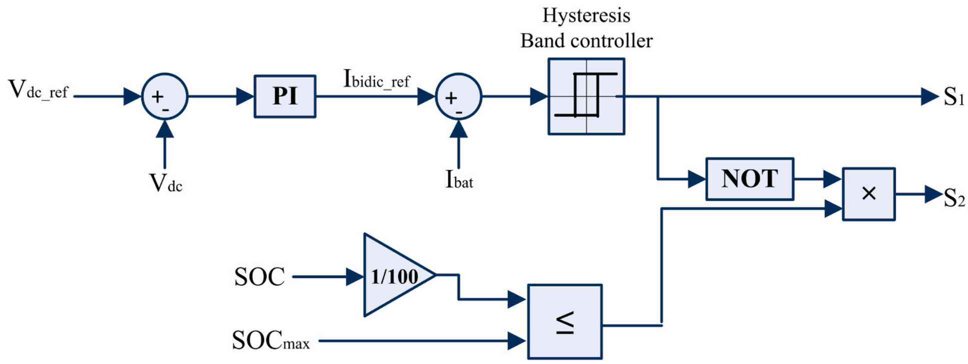

Fig. 4. Control strategy diagram of BIDIC converter.

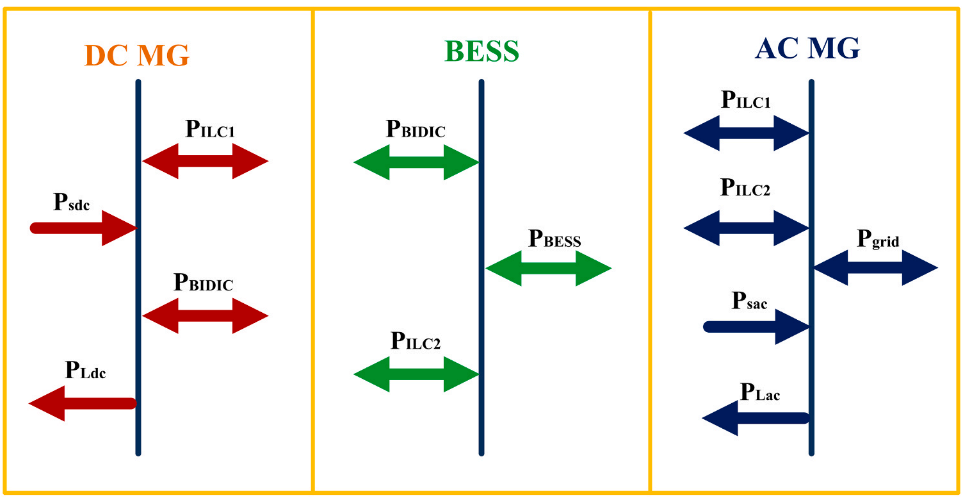

Fig. 5. Power flow of proposed AC/DC HMG.

2. Proposed controller of converters

The main components of hybrid microgrid are the ILC converters and ESD. The units used in the hybrid microgrid are outlined in Fig. 1.

2.1. Control of ILC1 and ILC2

The ILC1 is a device that performs bidirectional power sharing between DC and AC microgrids according to Fig. 1. This converter main task is forming the DCMG, in addition to the power sharing in the hybrid microgrid.

The ILC2 shown in Fig. 1 represents a power converter, connecting the ACMG to the ESD, and thus, operates as a backup to the grid. The ILC1 and ILC2 have similar structures and their performance is almost the same. The ESD provides the DC link of the ILC2, but the DCMG voltage is also the DC-link of the ILC1. Other control specifications have the same process as the ILC1.

The control strategy of both ILC1 and ILC2 is based on VSG. The VSG strategy is really appropriate for grid-connected and islanded modes. This strategy injects virtual inertia into the DG units and as a result, it guarantees the system stability in islanded and grid connected modes. Fig. 3 demonstrates the overall system of the VSG method for controlling ILC converters.

The mathematical model for the VSG module in Fig. 3 can be expressed as:

$$\begin{cases} P_{in} + D_p\left(\omega_m^* - \omega_m\right) - P_{out} = J\omega_m^* \dfrac{d\omega_m}{dt} \\ Q_{ref} + D_q(V^* - V) - Q_{out} = K\dfrac{dE}{dt} \\ \theta_m = \displaystyle\int \omega_m dt \end{cases} \tag{9}$$

where P_{in} shows the output power of the governor, Q_{ref} represents the reference reactive power, D_p is the virtual damping coefficient in per units, D_q is the $Q-V$ droop coefficient in per units, P_{out} and Q_{out} are the feedback values of real and reactive powers, J and K represent the inertia gains of the real and reactive power loops, ω_m^* and ω_m are the nominal and virtual angular velocities of the rotor, V^* and V represent the nominal and output voltage amplitudes, E is the voltage of the VSG, and θ_m is the virtual rotor angle.

The governor block in Fig. 3 is an $\omega-P$ droop controller, which can be expressed as:

$$P_{in} = P_{ref} - K_{gov}\frac{\omega_m - \omega_0}{\omega_0} \tag{10}$$

where P_{ref} is the reference real power, and K_{gov} represents the droop gain of the governor in per units.

2.2. BIDIC controller

The BIDIC is a grid-supporting converter that connects the ESD to the DCMG as illustrated in Fig. 1. Its responsibility is supporting power

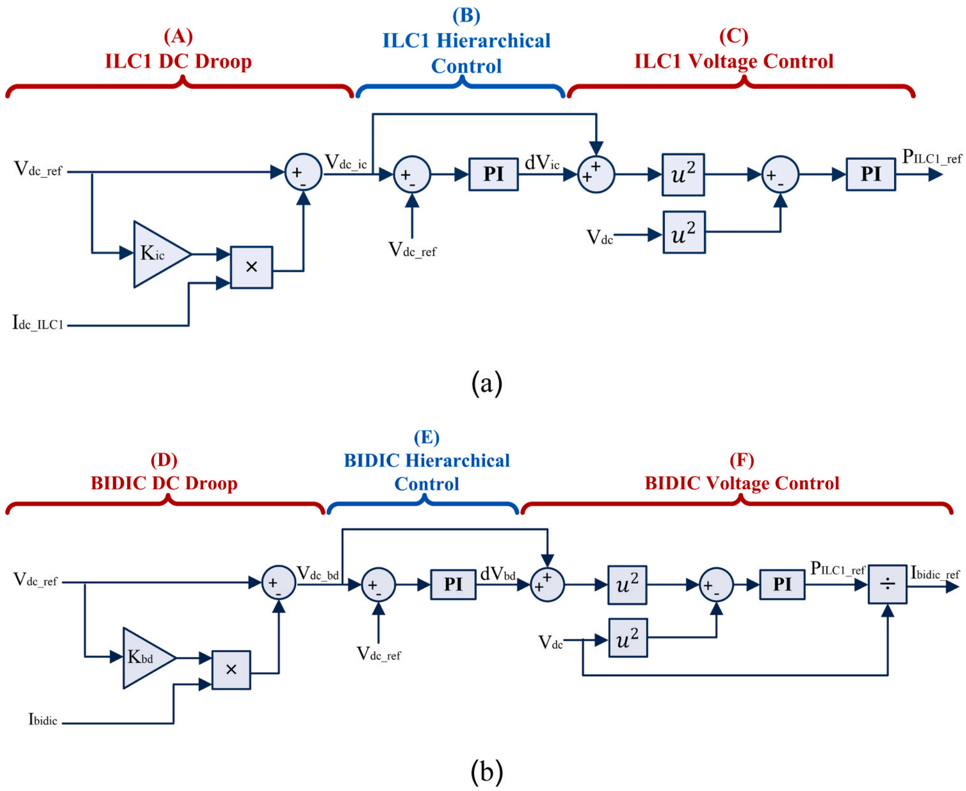

Fig. 6. Power management strategies for (a) ILC1 and (b) BIDIC.

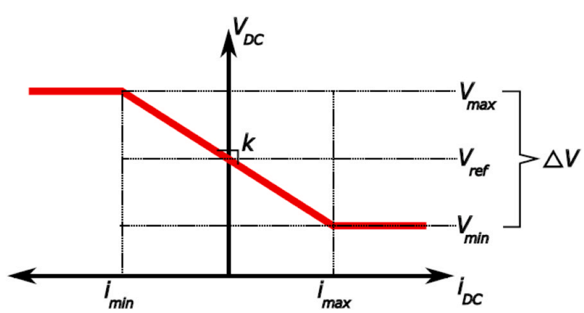

Fig. 7. Droop behavior characteristic [49].

sharing as well as regulating the DC-link voltage of the DCMG. The ESD and ILC2 are used along with the BIDIC. The BIDIC control scheme can be observed in Fig. 4.

This BIDIC has two controls including inner loop (current control) and outer loop (voltage control). The responsibility of this converter is to control the DC voltage of the DCMG with the support of the ESD. In addition, the mentioned bidirectional converter can exchange active power among AC and DC microgrids by the ILC2. This is done to support the ILC1 so that avoids the overcurrent occurrence in this converter.

The current loop shown in Fig. 4 obtains the reference current and calculates its deviation with the measured value. A hysteresis band controller is used after obtaining this error, so that generates the switching signals of switches S_1 and S_2. According to this figure, a battery charge controller is employed to prevent overcharging of the ESD by comparing the current SOC and $SOC_{\max}$.

2.3. Power management strategy

Based on what mentioned in the prior sections, a scheme is suggested for managing power by using the control strategies provided for the ILCs, improving the hybrid microgrid power sharing. This power management strategy considers power sharing, the reactive power compensation, as well as the voltage adjustment.

A. Voltage regulating and power sharing

The power flow in the hybrid microgrid is assessed for 3 main buses including the DCMG, battery energy storage system (BESS), and the ACMG, which can be observed in Fig. 5. The introduced energy management scheme considers the hybrid microgrid structure and operates by sharing the required power of each microgrid between the ESD and converters of the distribution system. Notably, the analysis is valid for both grid-connected and islanded modes. P_{grid} and Q_{grid} are considered to be zero in the islanded case. Moreover, the total consumption of the microgrid in both the DC and AC sides should not be the same as or higher than the total produced power of the two sides in the islanded mode.

According to the DCMG in Fig. 5, the power balance of the DCMG is presented as:

$$P_{DCMG} = P_{S_{dc}} + P_{L_{DC}} = P_{ILC1} + P_{bidic} \tag{11}$$

where P_{DCMG} is the DC microgrid power balance, and P_{ILC1} represents the real power of the ILC1.

The difference between powers of the DC sources and load is divided between the BIDIC and the ILC1. In addition to the power sharing capability, the energy management scheme can be established for regulating the DC link voltage of the DCMG. Therefore, energy management strategies are augmented to the control schemes of the BIDIC and the ILC based on Fig. 6 (Silveira et al., 2021).

In the DCMG, the droop control based power sharing method needs an additional voltage loop; they are shown in parts A and D in Fig. 6. The converters' power sharing is enabled by the DC droop controller based on the associated rated power and the allowed each converter voltage variations as can be seen in Fig. 7.

The droop control characteristics provided in Fig. 7 are consecutively written as (12) and (13) for the ILC1 and the BIDIC.

Table 2
AC/DC HMG system parameters.

Parameters	Values
HMG base power (P_{base})	$6kW$
Base voltage and frequency of AC side (V_{base},f_{base})	$220V - 60Hz$
Grid short-circuit capacity (S_{sc}) and X/R ratio	$5kVA - 0.04$
ILCs filter (R_{ilc}, L_{ilc})	1Ω, $1.5mH$
Virtual inertia coefficients of ILCs (J)	$1.5kg.m^2$
Damping coefficients of ILCs (D)	$17p.u$
Voltage controller gains of BIDIC converter	$k_p = 1.5$, $k_i = 25$
Input inductance of BIDIC converter	$L_B = 10mH$
Y/Y isolated transformer	$P_n = 5kVA, a = \frac{220V}{220V}$ $R_p = R_s = 0.16p.u$ $L_p = L_s = 0.04p.u$
Voltage and capacity of ion-lithium battery	$V_n = 325V$, $C_n = 6.3Ah$
Voltage and capacity of DCMG	$C_{DCMG} = 2250\,\mu F$, $V_{DC} = 410V$
DC-link voltage and capacity of ILC2	$C_{BESS} = 1125\,\mu F$, $V_{DC} = 350V$

$$V_{DC_{ic}} = V_{DC_{ref}} + K_{ic} I_{DC_{ILC1}} \Rightarrow \quad K_{ic} = \frac{V_{DC_{ref}}}{2P_{nom_{ilc1}}} \Delta V_{DC} \tag{12}$$

$$V_{DC_{bd}} = V_{DC_{ref}} - K_{bd} I_{bidic} \Rightarrow \quad K_{bd} = \frac{V_{DC_{ref}}}{2P_{nom_{bd}}} \Delta V_{DC} \tag{13}$$

where $V_{DC_{ref}}$ denotes the reference voltage of the DC link; $V_{DC_{ic}}$ and $V_{DC_{bd}}$ are the droop reference voltages of ILC1 and BIDIC; K_{ic} and K_{bd} are the DC droop constants of ILC1 and BIDIC; $P_{nom_{ilc1}}$ and $P_{nom_{bd}}$ are the rated powers of ILC1 and BIDIC; $I_{DC_{ILC1}}$ represents the DC current of the ILC1, and ΔV_{DC} shows the allowable voltage variation range.

The merits of droop method can be stated as no required communications among converters as well as its simple application. Therefore, the droop characteristics can respond to the DC link voltage deviation. This change can be modified using the secondary level of the hierarchical controller where a PI is applied to compensate the steady-state voltage error according to (14) and (15):

$$dV_{ic} = K_{pd_{ic}} \left(V_{DC_{ic}} - V_{DC_{ref}} \right) + K_{id_{ic}} \int \left(V_{DC_{ic}} - V_{DC_{ref}} \right) dt \tag{14}$$

$$dV_{bd} = K_{pd_{bd}} \left(V_{DC_{bd}} - V_{DC_{ref}} \right) + K_{id_{bd}} \int \left(V_{DC_{bd}} - V_{DC_{ref}} \right) dt \tag{15}$$

The hierarchical controllers of ILC1 and BIDIC are illustrated in Fig. 6, parts B and E. The modified reference droop voltage is the DC reference voltage according to parts C and F of Fig. 6.

Table 3
Operating condition of each event.

Events	1	2	3	4	5
Time duration (s)	**0–1**	**1–2**	**2–3**	**3–4**	**4–5**
Power flow condition	$P_{ACMG} = 0$ $P_{DCMG} = 0$	$P_{ACMG} > 0$ $P_{DCMG} > 0$	$P_{ACMG} > 0$ $P_{DCMG} < 0$	$P_{ACMG} < 0$ $P_{DCMG} < 0$	$P_{ACMG} < 0$ $P_{DCMG} > 0$

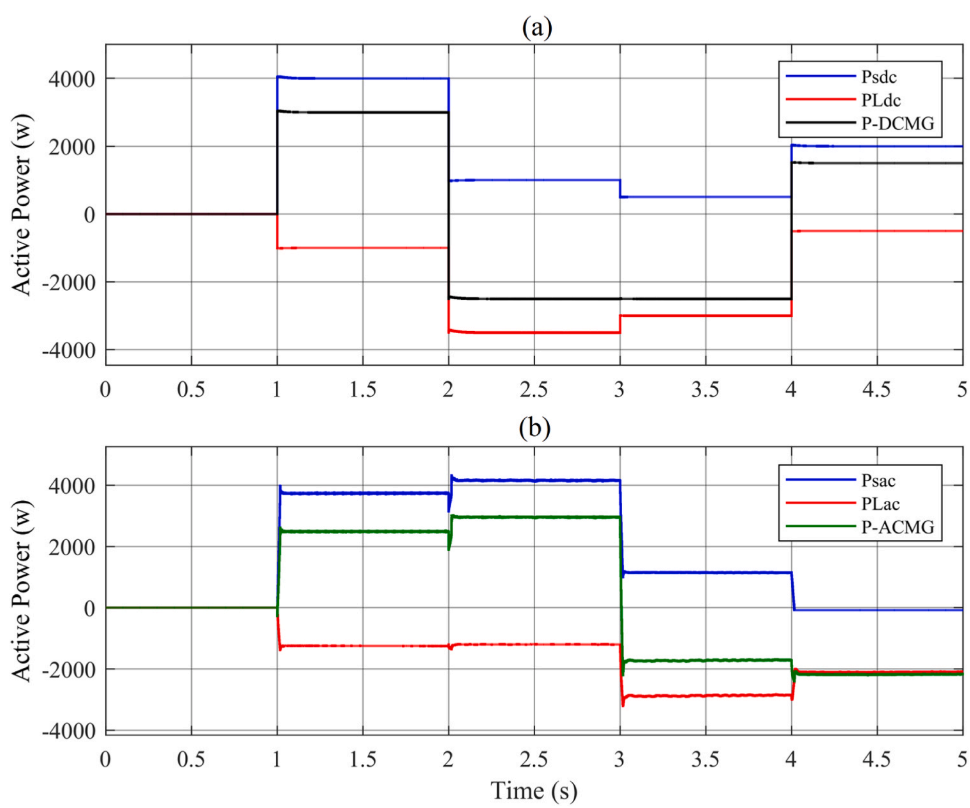

Fig. 8. Power flow profile of sources and loads. (a) DCMG, (b) ACMG.

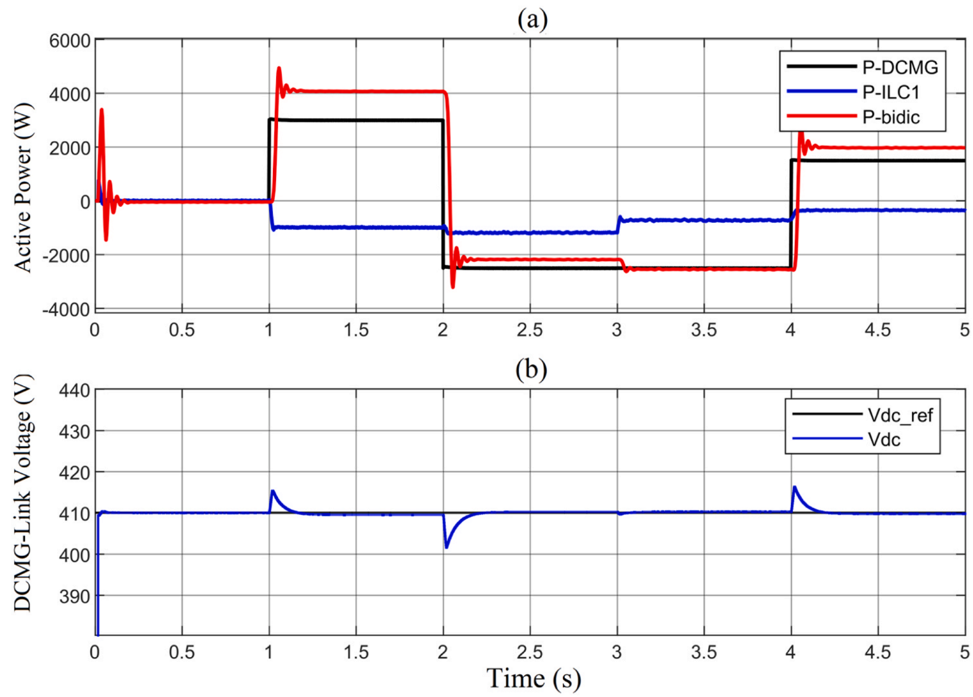

Fig. 9. DCMG simulation results. (a) Power sharing, (b) DC-Link voltage with deviation correction by hierarchical control.

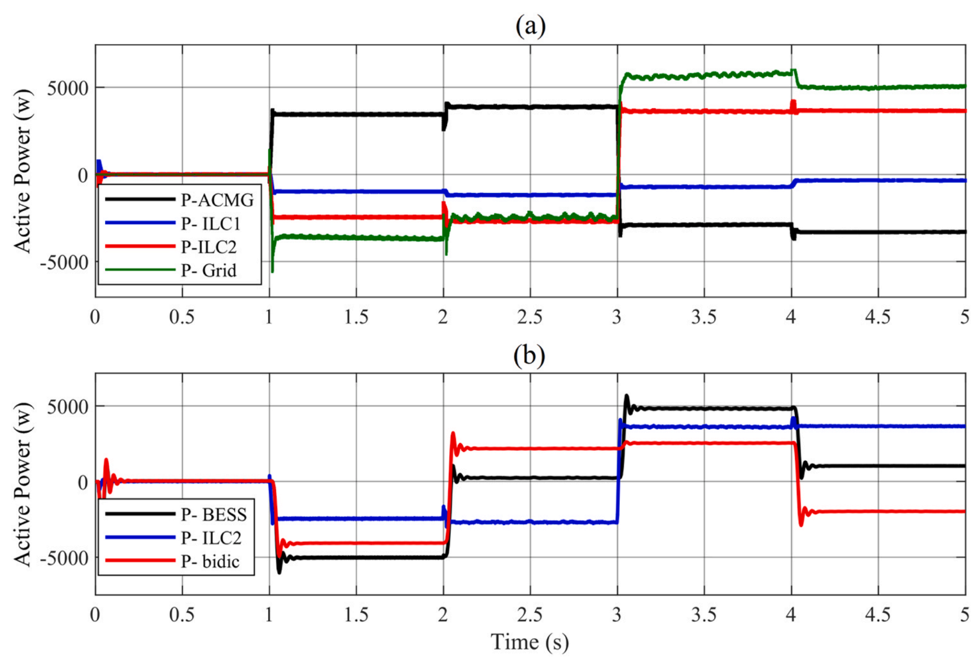

Fig. 10. ACMG simulation results. (a) ACMG power sharing, (b) TSILC power flow.

In the ACMG, the behavior of transmitting the real power can be tracked according to Fig. 5, the ACMG part. A power profile is considered for the sources and loads, and the power of ILC1 can be expressed through the DCMG droop controller. Thus, in the ACMG, sharing power among the power grid and the ILC2 is realized according to the following relationships:

$$P_{ACMG} = P_{S_{ac}} + P_{L_{ac}} \tag{16}$$

$$P_{grid} = P_{ACMG} + P_{ILC1} + P_{ILC2} \tag{17}$$

The considered power sharing strategy for ACMG management is the master-slave controller; ILC2 is considered as the master unit. In this method, the reference power of ILC2 is selected based on the demanded power management scheme. The aim of this power management system is to share power between the power grid and the ESD.Top of Form Consequently, the reference real power of ILC2 can be stated as (18) in which the remaining power is transferred to the power system.

$$P_{ILC2_{ref}} = \frac{1}{2}(P_{ACMG} + P_{ILC1}) \tag{18}$$

where $P_{ILC2_{ref}}$ is the reference active power of ILC2.

On the other hand, the behavior of the power sharing of the TSILC

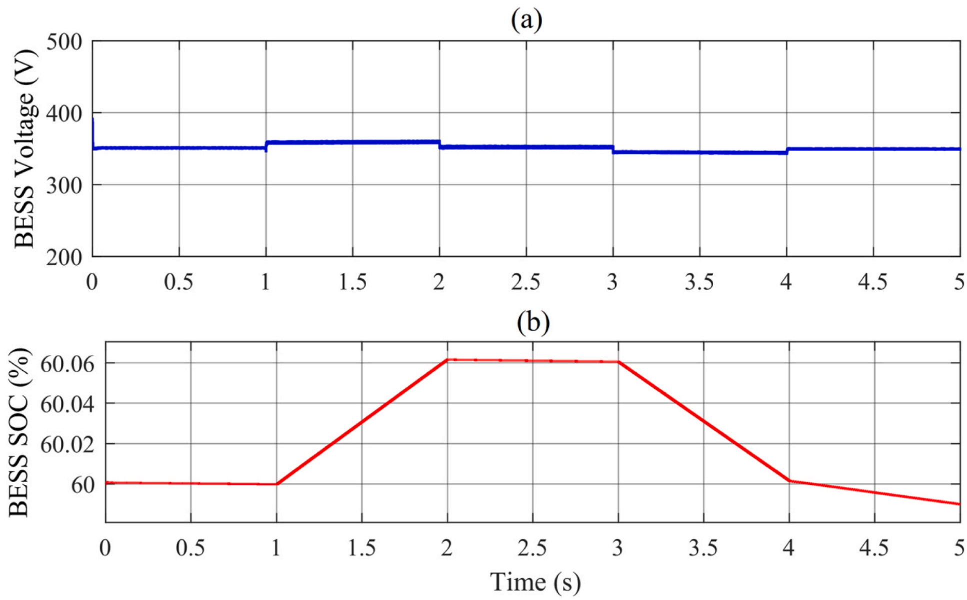

Fig. 11. BESS outputs under power flow in TSILC. (a) Battery voltage, (b) Battery SOC.

with ESD can be tracked based on Fig. 5, the BESS part. The battery bank provides balance among powers of bidirectional converters and ILC2 according to (19). This enables simultaneously managing AC and DC MGs through TSILC, leading to a centralized ESD based support.

$$P_{bess} = P_{bidic} + P_{ILC2} \tag{19}$$

where P_{bess} is the power of the battery bank.

B. Compensation of the reactive power

A reduction in the power factor of the PCC bus is resulted by reactive power consumption of sources and loads. The behavior of the reactive power sharing in the analyzed hybrid microgrid can be tracked as Fig. 5, the AC MG part. This behavior is expressed by (20). Consequently, the proposed power management system utilizes reactive power compensation to support the power system in grid-connected mode.

$$Q_{grid} = Q_{Sac} + Q_{Lac} + Q_{ILC1} + Q_{ILC2} \tag{20}$$

The ILC2 is a converter employed to compensate the reactive power. To do so, a master-slave strategy is implemented to define the reference reactive power of ILC1 as the total reactive power of sources and loads. The goal is to decrease the reactive power of Q_{grid} to zero according to the (22). The ILC1 operates with no reactive power.

$$Q_{grid} = Q_{ILC2} = 0 \tag{21}$$

$$Q_{ILC1\_ref} = -\left(Q_{Sac} + Q_{Lac} + Q_{grid}\right) \tag{22}$$

where $Q_{ILC1-ref}$ is the reference reactive power of the ILC1.

3. Simulation results

A typical hybrid AC/DC MG as Fig. 1 is simulated in MATLAB/Simulink, so that evaluates the operation of the introduced power management scheme implemented in the improved ILCs. This assessment includes the analysis of power sharing and reactive power compensation in the PCC bus as well as the regulation of the DC link voltage of the DCMG, and is performed in two grid connected and islanded modes. The maximum nominal capacities of loads and resources are considered up to 4.5 kW. The ILC1 and ILC2 are each developed to have 3.5 kW rated power; this enables proper operation of the power management scheme. The values of inductors, capacitors, and converter controllers are applied according to Table 2.

The parameters of PI in the hierarchical controller are considered as $K_{pILC1} = K_{pbidic} = 0.098$ and $K_{iILC1} = K_{ibidic} = 2.85$. The discretization of controllers is performed by the Tustin strategy, having 20 kHz sampling frequency (Tian et al., 2017). The switching of the BIDIC is performed through the hysteresis band control, while the switching of both ILC1 and ILC2 is under 15 kHz frequency and performed by two-level SPWM modulation. The DC link reference voltage in the DCMG is 410 V. The acceptable voltage variation of the hierarchical droop controller is 20 V, in 400–420 V range. The output voltage of the battery bank (BESS) is 350 V.

Analysis of the introduced VSG based HMG is realized under islanded and grid-connected modes, in four different operational modes, including all bidirectional power transmission among AC and DC MGs. Thus, five power changes occur in the microgrids during the simulation. These changes are described in Table 3.

Because of the burden of calculation arising from the complexity of the system, the behavior of the proposed hybrid microgrid is simulated in 5 s. During 0–1 s, occurring a transition is observed; the steady state DC link voltage is reached, and in this state, no power flows in the microgrid. Hence, the operational conditions occur for the periods given in Table 2. According to these cases, the DC and AC loads and resources associated profiles in the respective microgrids are defined, as can be observed in Figs. 8a and 8b.

The load related power sharing profiles and that of power suppliers demonstrated in Figs. 8a and 8b are used in the analysis of voltage adjustment, power sharing, and reactive power compensation for islanded and grid-connected modes. Each analysis is discussed in detail in the following sections.

3.1. Operation of proposed HMG under grid-connected mode

Based on Fig. 1, the proposed microgrid is linked to the three-phase power system by a circuit breaker. Therefore, the power system can support the hybrid microgrid in these conditions by exchanging active and reactive powers.

A. Voltage regulating and power sharing

The evaluation of sharing power in DC and AC microgrids is conducted using the methods provided in Section 3. In the DCMG, a DC droop controller is utilized for ILC1 and BIDIC. As a result, as shown in Fig. 9, the disparity between powers of the DC load and the DC source is divided among the converters.

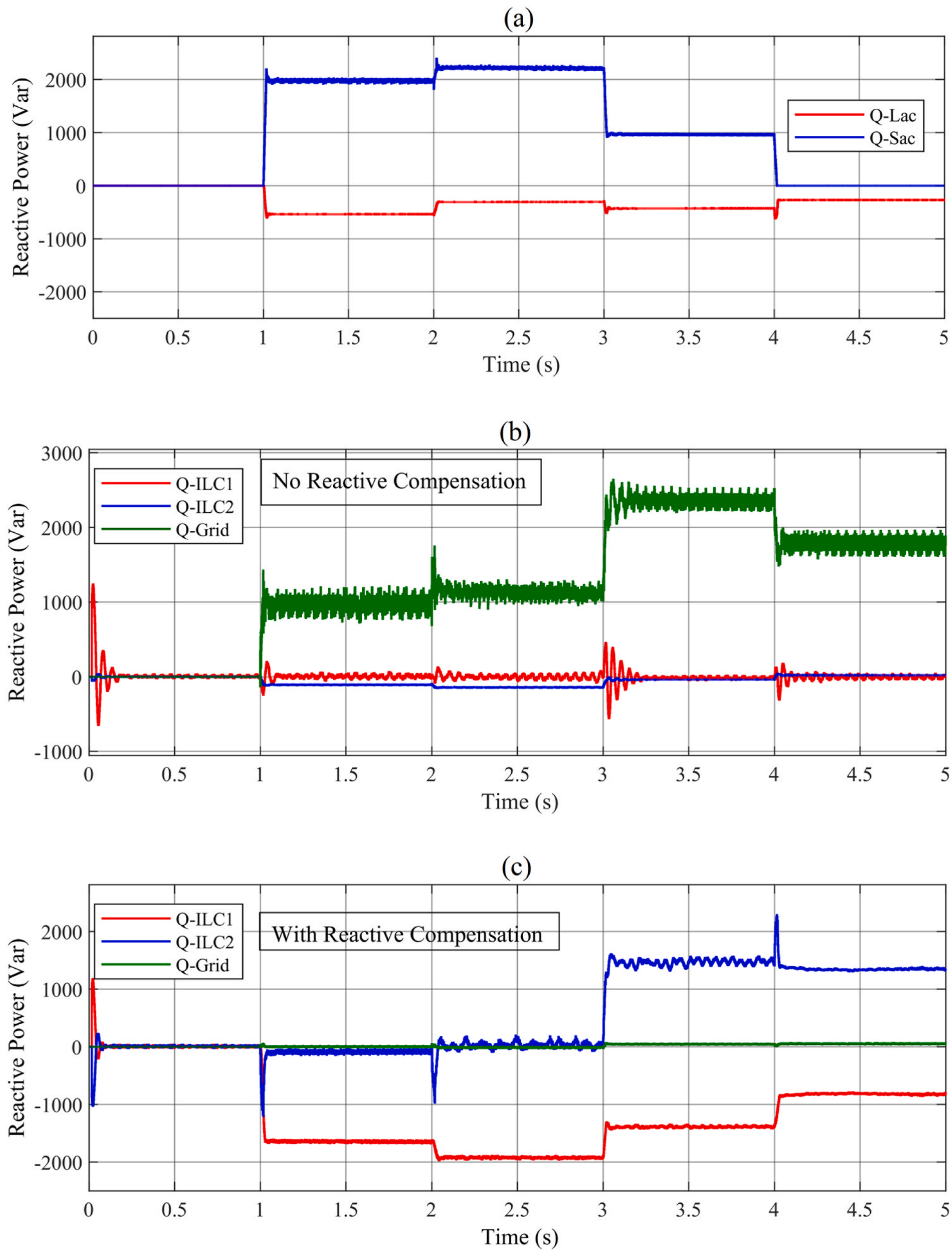

Fig. 12. Analyzing the reactive power flow under grid-connected mode. (a) Reactive power profile of ACMG, (b) reactive power of converters regardless of compensating, (c) reactive power of converters with compensation.

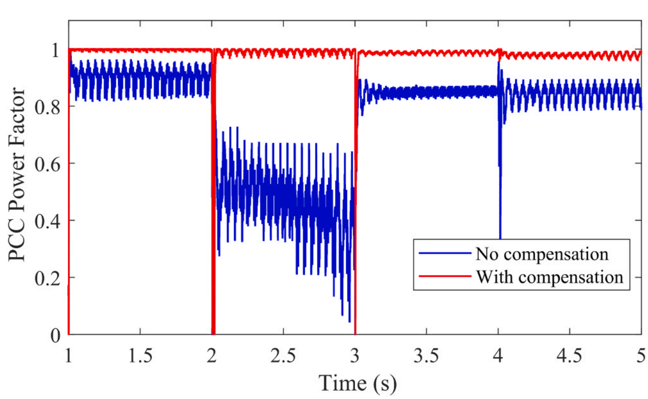

Fig. 13. Power factor of PCC bus.

According to Fig. 9a, the DCMG power balance is represented by the the black curve; this tracks the operational cases of Table 2. Since the rated powers of the BIDIC and the ILC1 are not similar, the balance in powers is not equally divided among them. Consequently, during the period of 1 −4 s, ILC1 and BIDIC receive unequal power from the DC link of the islanded microgrid. During 1–2 s, the BIDIC injects 4 kW from the DC microgrid; since this power is more than the output of the DC microgrid (3 kW), the rest of it, i.e. 1 kW, is transferred through the ILC1 to the DCMG and then to the BIDIC. Between 2 and 4 s, the DCMG required power, i.e. 2.5 kW, is transferred to the DC through ILC1 and BIDIC. During 4–5 s, 2 kW is absorbed by the ACMG through the BIDIC, and its additional amount (about 0.5 kW) is transferred to the DCMG and then the BIDIC by the ILC1. It should be noted that the power sharing of the ILC1 is directly performed in the ACMG, while the power sharing of the BIDIC depends on the battery bank. The reason for the high power exchange by the BIDIC compared to the ILC1 is the support

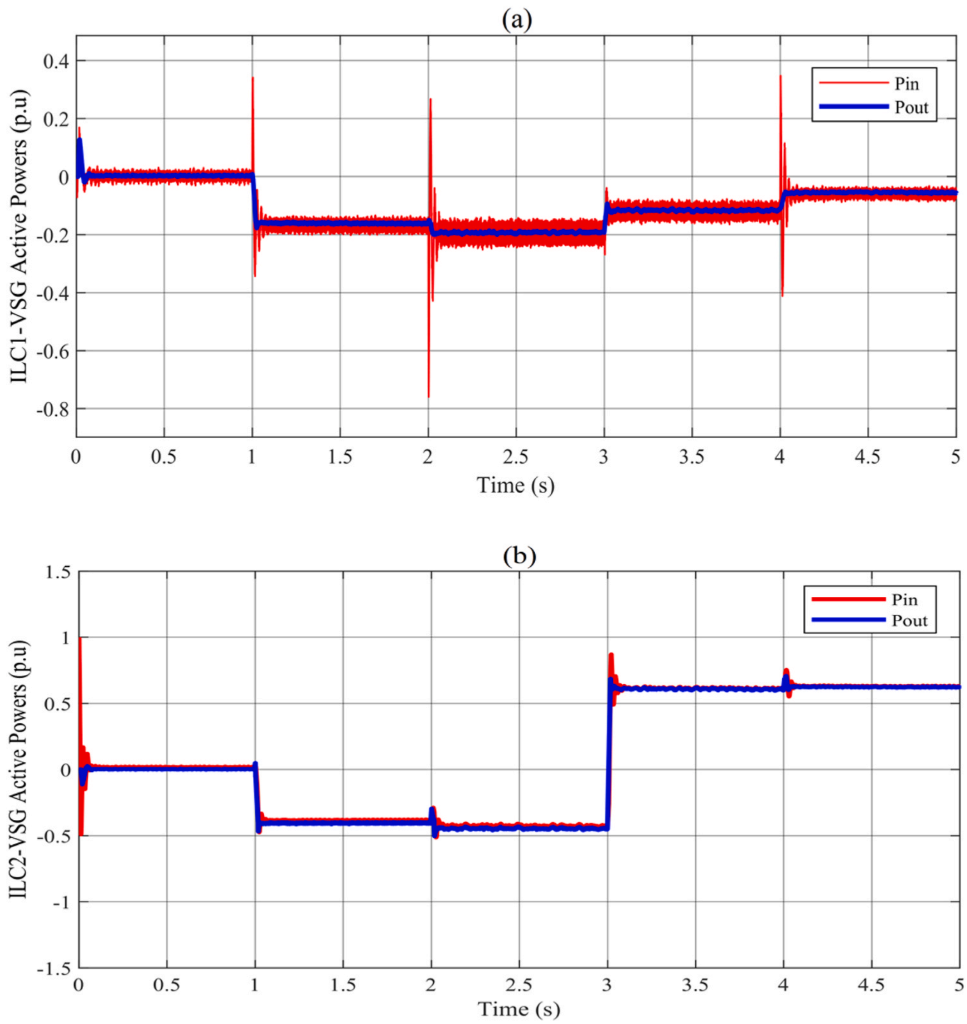

Fig. 14. Input and output powers of VSGs under various profile of sources and loads in grid-connected mode. (a) ILC1, (b) ILC2.

provided by the BESS.

The PI compensator as well as the hierarchical controller are implemented to adjust the DC link voltage deviation. The voltage adjustment in the DCMG can be observed in Fig. 9b. It can be clearly observed that the hierarchical control can correct the voltage correction during the steady state and the DC voltage error is almost zero.

In the ACMG, the power is divided between AC sources and loads, the ILC1, the ILC2, and the distribution system. The power sharing of the ILC1 leads to sharing power in the DCMG, however, the profile presented in Fig. 8b determines the power balance among loads and sources of the ACMG.

The equal power sharing among the ILC2 and the power system is performed, in which the ILC2 implements the master-slave controller, producing the reference power of the converter as (18).

The power sharing of the ACMG is depicted in Fig. 10a. By evaluating each of the power changes, it is possible to confirm the power sharing between ILC2 and the distribution system for each operational condition. At 1–2 s, the powers of the AC resource and the ILC1 are transferred to the AC microgrid by absorbing the energy of the ILC2 and the power grid. During 2–3 s, the ILC1 absorbs the AC power according to the conditions of the DCMG. Thus, the AC source continues to produce the power required by ILC1, and as a result, ILC2 and the power grid are still in absorbing power. From 3–4 s, the AC loads are dominant and the ILC2 and the grid need to inject power to the ACMG. During 4–5 s, the ILC1 starts absorbing power, but the operational conditions remain the same. The ILC1 absorbs power from the ACMG and delivers it to the ACMG in the entire simulation process.

The BIDIC and ILC2 are converters that consist of a TSILC and a hybrid microgrid interface converter that acts as an energy storage source. Therefore, there is a power sharing on the BESS side due to the behavior of the BIDIC and ILC2, as shown in Fig. 10b. It is clear that in this condition, the battery should inject or absorb energy into or to both AC and DC MGs, which is shown in the time intervals of 1–2 - and 3–4 s. During 2 –4 s, the each MG supports the other MG's power sharing. As a result, the battery required power is mitigated here.

The status of the BESS is illustrated in Fig. 11 in terms of battery voltage and its SOC under TSILC power sharing condition. According to Fig. 11a, it seems that the hierarchical strategy based on the droop controller in the BIDIC causes the battery voltage to vary around the nominal value of 350 V depending on the state of converter power transmission. Fig. 11b shows the SOC of the BESS under TSILC's load sharing. It can be seen that between 1 and 2 s, the battery is charged through BIDIC and ILC2. During 2–3 s, the power of the aforementioned converters is equal and opposite to each other, and the SOC of the battery remains constant. From 3–4 s, the battery is discharged through two BIDIC and ILC2. Finally, the battery discharge rate decreases between 4 and 5 s.

B. Reactive power compensation under grid-connected mode

In the ACMG, rotating machine like induction motor is frequently utilized, requiring significant consumption of the reactive power. Therefore, the power factor of the distribution system may be influenced. The harmonic distortion is the other aspect affecting the power

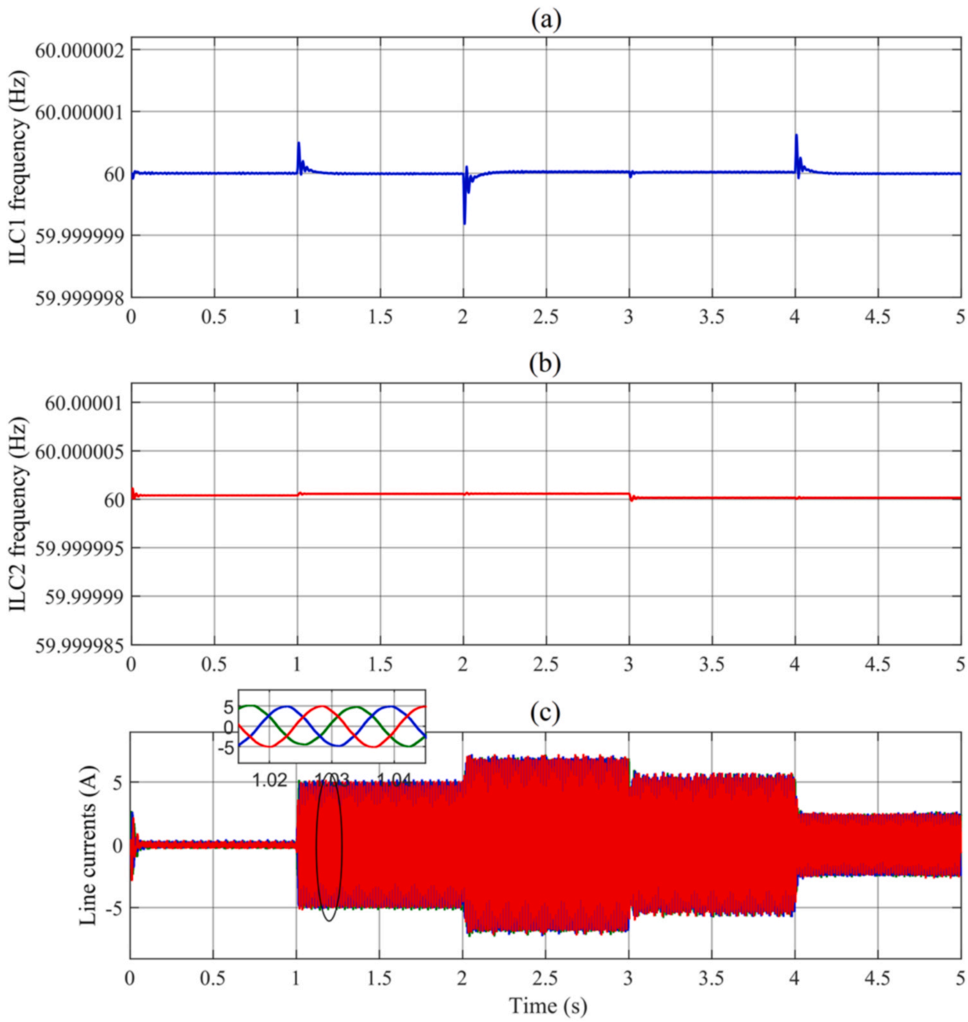

Fig. 15. VSGs outputs under various profile of sources and loads in grid-connected mode. (a) ILC1 frequency, (b) ILC2 frequency, (c) ACMG line currents.

factor. The reactive power profile for AC loads and sources is generated for the effect of reactive power analysis in the hybrid microgrid and is presented in Fig. 12a. The AC source injected reactive power can be observed as the blue curve in Fig. 12a; the red one demonstrates the AC load absorbed reactive power. This reactive power profile directly impacts the power factor of the PCC bus in the ACMG according to Fig. 12b.

Fig. 12b represents the reactive power in the PCC bus of the distribution system while there is no compensation for the reactive power. As a result, the ILC1 and ILC2 are performed by the power factor of 1, which can be observed in the blue and red curves. It seems according to the green curve that the reactive power is fed to the distribution system during 1–5 s, and its values are different for different time intervals depending on the additional reactive power produced by the ACMG.

The compensation of the reactive power is demonstrated in Fig. 12c. In the green curves of Fig. 12c, the PCC reactive power of the distribution system can be observed. The changes in the green curve are the transient response, occurring while changing the operational conditions. After this transient state, the reactive power approaches zero when it reaches the steady state. The red curve gives the reactive power of ILC1, this is used for the AC load and source respective reactive profile compensation as (22). The ILC2 is implemented to compensate a part of the system's reactive power; this is presented by the blue curve in Fig. 12c.

The performance of reactive power compensation is illustrated in Fig. 13, comparing the power factor of system in the case of with and without reactive power compensation. By considering the compensator, the obtained power factor is presented by the red curve in Fig. 13, while the blue curve presents the results of without compensation. The distribution system and PCC bus associated power factor using the proposed compensation method equals 1 in all operational intervals of the microgrid (red curve), while without the reactive power compensation, the system's power factor is unfavorable and even between 1 and 2 s reaches to a 0.4 lagging power factor (blue curve).

As a result, the reactive power sharing enhancement in compensation mode is evident. The transient states occur during the changes in operational conditions at the beginning of 2 −3 s. These transient states happen because of the variations in the reactive power direction.

C. Virtual synchronous generator operation in ILCs

In this section, the behavior of VSG controllers in the ILCs is evaluated in the grid connected mode. The reference input and output powers of VSGs of both ILC1 and ILC2 are presented in Fig. 14. The input power of VSGs is the same as the output of the governor block. Here, the governor transfer function is expressed as:

$$G_{gov}(s) = \frac{0.5}{0.01s + 1} \tag{23}$$

Based on Fig. 14, it is observed that the output powers of both VSGs match their reference powers and these changes are caused by the variations in their output frequency. The frequency variation of each VSG is illustrated in Fig. 15. The slight frequency change in each VSG is accompanied by a significant active power exchange in the ILCs. By comparing Fig. 14a and Fig. 15a as well as Fig. 14b and Fig. 15b, the

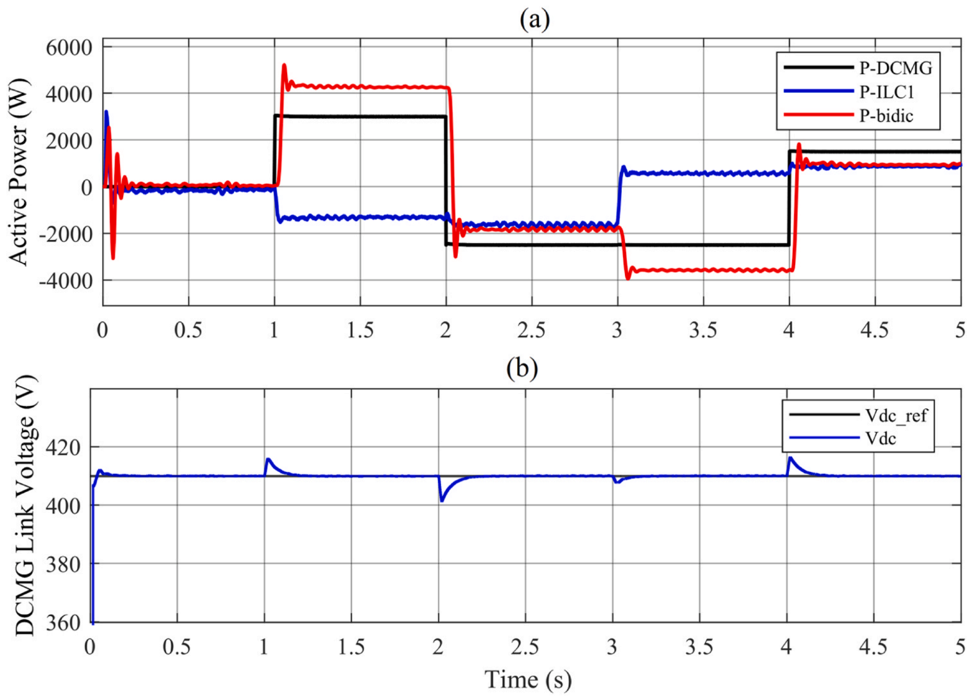

Fig. 16. DCMG simulation results under islanded mode. (a) Power sharing, (b) corrected deviation of DC-Link voltage by hierarchical controller.

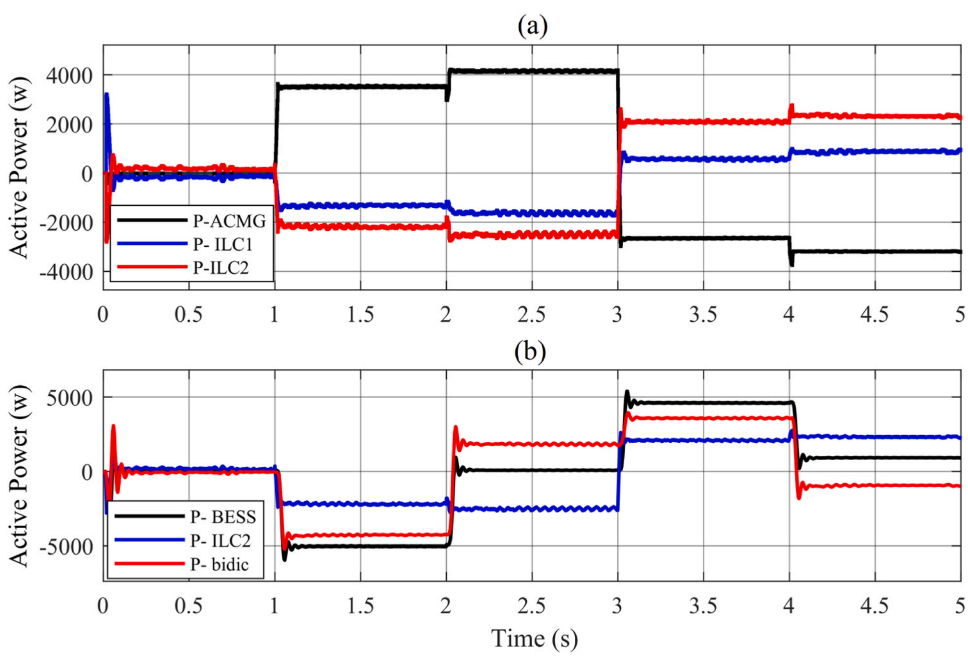

Fig. 17. ACMG simulation results under islanded mode. (a) ACMG power sharing, (b) TSILC power flow.

active power absorption (with a negative sign) of each ILC corresponds to a slight increase in the VSG frequency, and adversely, their active power injection (with a positive sign) corresponds to a slight decrease in VSG frequency. The three-phase current passing through the AC feeder in the ACMG during its power changes is shown in Fig. 15c which can be seen that it is sinusoid.

3.2. Operation of proposed HMG under Islanded mode

To confirm the capability of the proposed method, the hybrid AC/DC MGs including the profiles of the load and resources presented in Fig. 8 is simulated in the islanded mode. In these conditions, the microgrid is isolated from the distribution system by a circuit breaker, and as a result, there is no frequency and voltage support by the system. Besides, the islanded hybrid microgrid includes the same settings of the grid connected mode, and no change is observed in the controllers and their values. Thus, the proposed microgrid should allow controlling the frequency and voltage, as well as power sharing without the distribution system's participation.

A. Voltage regulating and power sharing

Similar to the previous assessment, as illustrated in Fig. 16a, the power disparity among the DC load and source can be divided by the

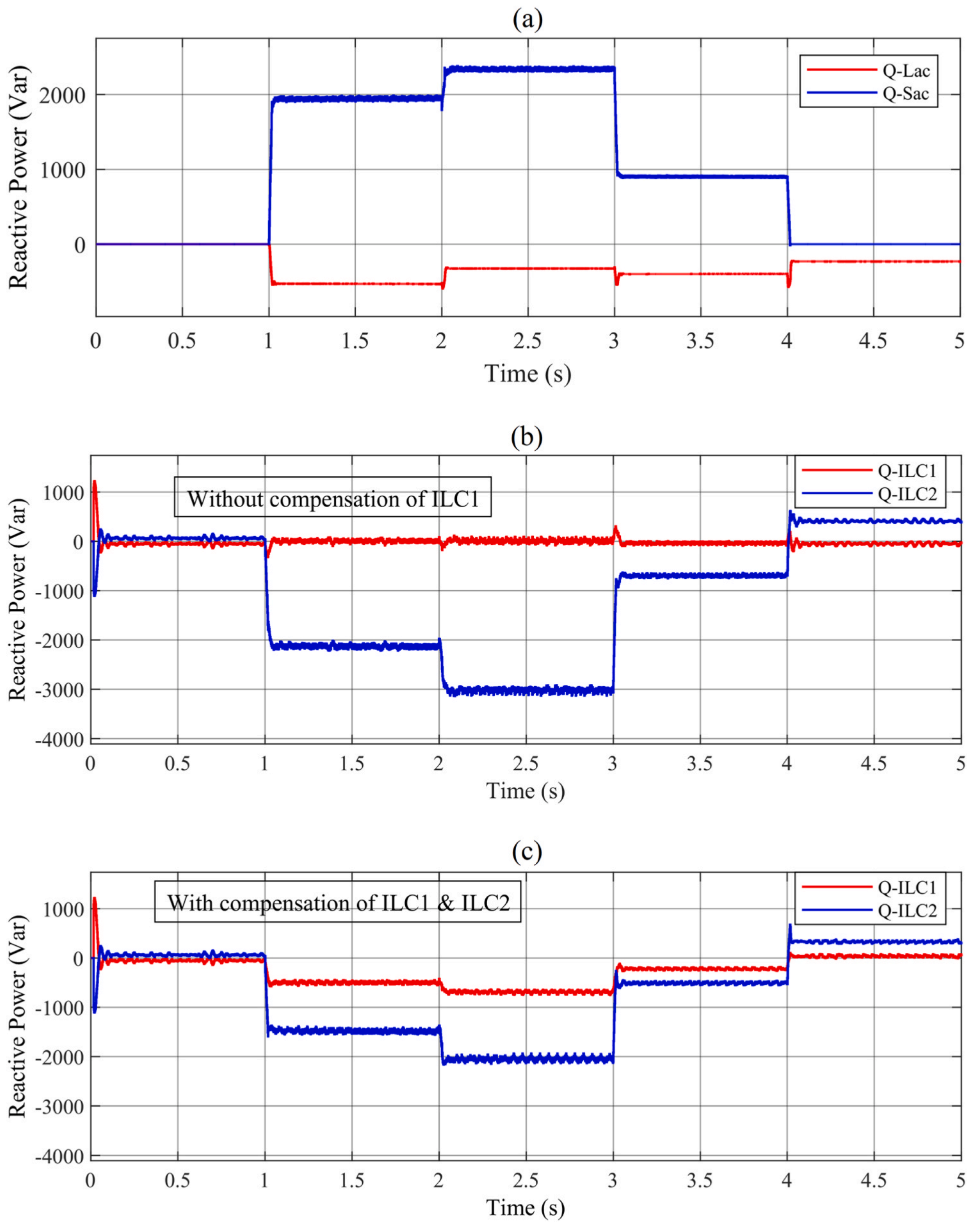

Fig. 18. Reactive power compensation under islanded mode. (a) Reactive power profile of ACMG, (b) reactive power compensation with only ILC2, (c) reactive power compensation with both ILC1 and ILC2.

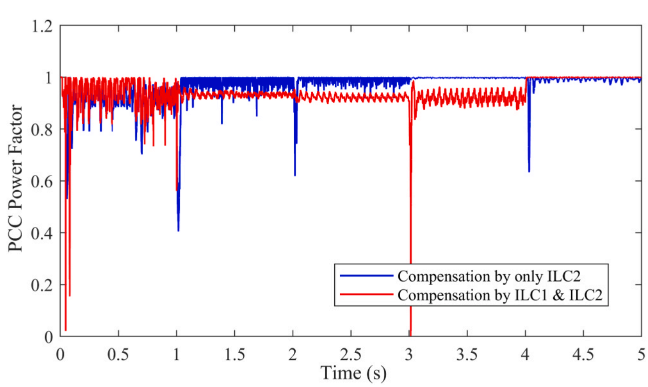

Fig. 19. Power factor of PCC bus under islanded mode.

converters. During 1 −2 s, the BIDIC receives about 4.2 kW from the DCMG and delivers it to the ACMG, 3 kW of it is absorbed by the DCMG sources, and the remaining, i.e. the ILC1's 1.2 kW produced power is absorbed by in the ACMG and delivered to the DC link.

From 2 s to 3 s, both converters absorb or inject the real power from or to the ACMG equally. During 3–4 s, the BIDIC injects about 1 kW more than the power required by the DCMG, and this excess amount is absorbed by the ILC1 and given to the ACMG. Eventually, during 4–5 s, the produced power of the DCMG which is about 1.8 kW is jointly absorbed by two converters and delivered to the ACMG. According to Fig. 16b, the DC link voltage of the DCMG is able to accurately follow its reference by the proposed modified hierarchical control method and stabilize it at 410 V after each power change.

Fig. 17 depicts the power sharing for the ACMG in islanded mode. Based on Fig. 17a, it seems that the ILC1 power is doubled in the grid connected mode without the support of the power grid. In these conditions, the ILC1 and ILC2 absorb almost the same amount of real power from the ACMG between 1 and 3 s. Moreover, both converters inject the

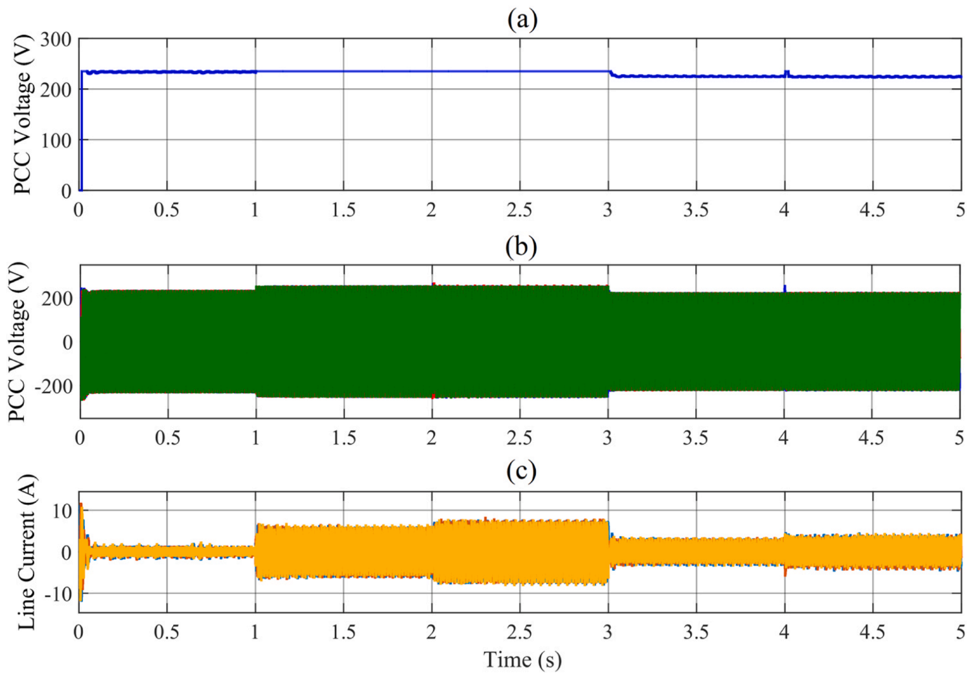

Fig. 20. Voltages and currents of PCC bus during reactive power compensation by ILC1 and ILC2. (a) RMS voltage, (b) three phase voltages, (c) three phsase currents.

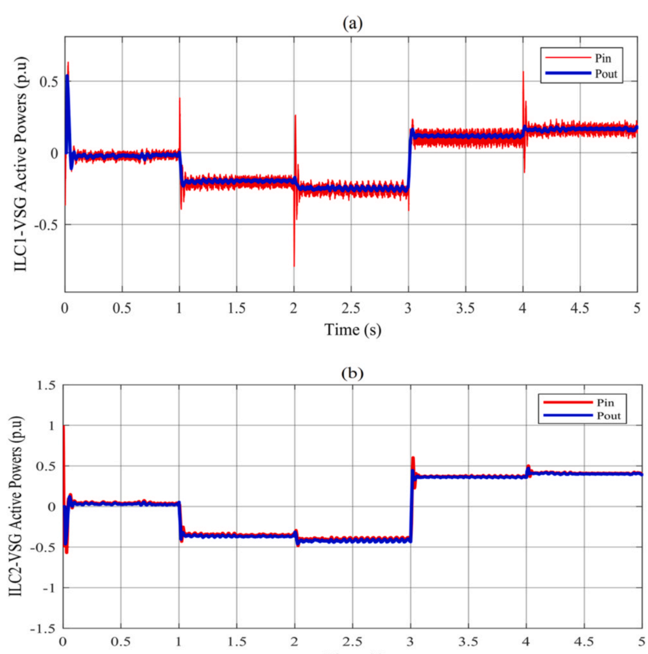

Fig. 21. Input and output powers of VSGs under various profile of sources and loads islanded mode. (a) ILC1, (b) ILC2.

real power into the AC MG during 3–4 s. It should be noted that according to (17), the algebraic sum of the powers of both ILC1 and ILC2 equals the ACMG required power (black curve). In addition, $P_{grid} = 0$ in the islanded mode.

Fig. 17b demonstrates the active power sharing of the TSILC in islanded conditions. According to (19), the algebraic sum of the active powers of BIDIC and ILC2 is equal to the BESS power. Therefore, according to Fig. 17b, between 1 and 2 s, the ILC2 and the BIDIC provide the required power to charge the BESS battery through the ACMG as well as the DCMG, respectively. During 2–3 s, the power of the two converters complements each other, and therefore, the power exchanged with BESS would be zero. The BESS is discharged about 4.5 kW during 3–4 s through the two mentioned converters. Lastly, the BESS is discharged around 1 kW between 4 and 5 s. It should be remembered that the SOC curve of the BESS module is the same as Fig. 11 in this condition.

A. Reactive power compensation under islanded mode

It is necessary to regulate the voltage and control the reactive power of the PCC bus of the ACMG in the islanded mode and without frequency and voltage support from the power grid. Therefore, here, both ILC1 and ILC2 must be able to exchange reactive power and maintain the AC side voltage at the nominal value. If only one of the converters (for example only ILC2) compensates the reactive power of the entire microgrid, then the reactive power sharing in the entire ACMG would not be uniform and the power factor of the PCC bus is unfavorable.

Fig. 18 represents the profile of ACMG sources and loads. In Fig. 18b, the compensation of the reactive power is conducted exclusively through the ILC2, the ILC1 does not participate in the reactive power compensation and its value is zero during the simulation process. The effect of this mode on the power factor is shown in Fig. 19. Despite the fact that the power factor of the PCC bus equals 1 during 3–5 s and is acceptable, it is severely distorted during 1–2 s and is undesirable.

Fig. 18c shows the reactive power compensation with the participation of both ILC1 and ILC2. Under these conditions, the reference reactive power of ILC1 is adjusted to compensate 35 % of the AC microgrid reactive power and the rest of the compensation (65 %) is performed through the ILC2. According to Fig. 18c, the ratio of reactive power compensation by the ILC1 is almost one third (35 %) of that of ILC2 in the entire compensation period from 1 to 5 s. This improves the power factor and effective power sharing in all loading ranges. Thus, the effect of reactive power sharing between ILC1 and ILC2 can be observed in Fig. 19.

Fig. 20 provides the waveforms of voltage and current at the PCC bus by considering the reactive power compensation. According to Fig. 20a and Fig. 20b, during the ACMG loading conditions, the PCC rms voltage is 235 V during 0–3 s, while in the range of 3–5 s, this voltage decreases to 225 V due to the compensation of the reactive power. A 5 % change in the voltage range can be acceptable for an islanded microgrid with the aim of producing a value above 0.9 lagging power factor. In Fig. 20c, the waveform related to the current passing through the AC feeder is

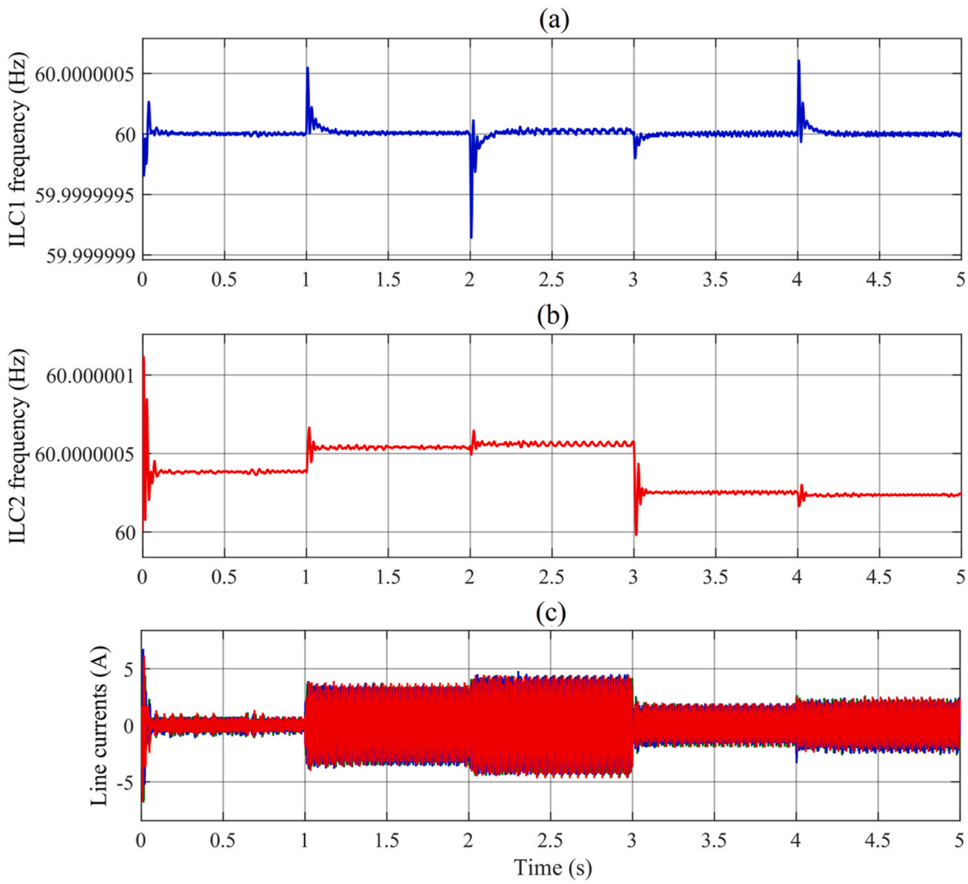

Fig. 22. VSGs outputs under various profile of sources and loads in islanded mode. (a) ILC1 frequency, (b) ILC2 frequency, (c) ACMG line currents.

represented under the load and source profiles.

C. Virtual synchronous generator operation in ILCs during islanded mode

The performance of microgrid VSGs in the absence of grid frequency support is investigated in the islanded mode similar to the grid-connected mode. In Fig. 21, it is clear that the input and output powers are matched in both VSGs of the ILC1 and ILC2. According to Fig. 21a, the active power contribution of the ILC1 increases almost twice in the islanded mode compared to the grid connected mode, while that of ILC2 remains almost stable (Fig. 21b).

The simulation results related to the frequency of VSGs are presented in Fig. 22. The slight frequency change in each VSG is accompanied by a significant active power exchange in the ILCs. By comparing Fig. 21a and Fig. 22a as well as Fig. 21b and Fig. 22b, the active power absorption (with a negative sign) of each ILC leads to a slight increase in the VSG frequency, and inversely, their active power injection (with a negative sign positive) is accompanied by a slight decrease in VSG frequency. The three-phase current passing through the AC feeder in the ACMG under its power changes is demonstrated in Fig. 22c.

4. Conclusion

In this work, a power management strategy in presence of an energy storage system was proposed for the modified structure of VSG based ILCs in order to produce virtual inertia in the DGs of hybrid AC/DC MGs. The aim of this strategy was to improve the power sharing of the grid-connected and islanded hybrid AC/DC MG. The power factor was modified using a reactive power compensator which was placed in the structure of VSG based ILCs. As a result, the AC loads' absorbed reactive power was well compensated. By implementing this method and reactive power compensation, the amplitudes of three-phase PCC voltages were also kept constant. A hierarchical control was proposed to attain power sharing and regulate the DC link voltage. The performance of the introduced power management system and the structure of the modified ILCs were evaluated and confirmed by simulations in MATLAB/Simulink. The following important conclusions can be stated:

- By using the introduced compensation scheme, the power factor of the distribution grid and PCC bus was 1 in the grid connected mode and in all operational intervals of the microgrid, while without reactive power compensation, the system power factor is unfavorable and even reached 0.4 lagging power factor in some moments.
- By the reactive power sharing between ILC1 and ILC2 in the islanded mode, it was possible to enhance the power factor and effective power sharing in all loading ranges. So the power factor of the PCC bus in the islanded mode was always above 0.92 lagging power factor under the participation of both ILC1 and ILC2.
- A power sharing strategy was also proposed based on the droop controller in the DCMG and master-slave controller in the ACMG, which were able to adjust the needed power sharing between the power grid and the ESD. This accurate power sharing reduced the power grid injected energy and prevented the consumption concentration on the battery system.
- Satisfactory performance of the voltage adjustment in the DC subgrid was attained through hierarchical control by modifying the droop method resulted voltage variations.

Despite the various advantages of the existing proposed scheme, there are some limitations such as resilience of microgrid against short circuit faults, cyber-attacks, and low voltage ride through (LVRT) capability in the present work. In the future works, small signal stability investigation, uncertainty study, hardware implementation in the form of hardware in loop (HIL) and estimation of the inertia constant (H) and damping coefficient (D) of the VSG controller will be conducted. In addition, a new control structure will be added to controller in level1 to compensate unbalanced voltage of PCC.

CRediT authorship contribution statement

Hayder Oleiwi Shami: Writing – review & editing, Writing – original draft, Visualization, Validation, Formal analysis, Data curation, Conceptualization. **Ali Basem:** Validation, Supervision, Software, Resources. **Ameer H. Al-Rubaye:** Writing – original draft, Visualization, Validation, Formal analysis. **Kiomars Sabzevari:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

Akbari, Ehsan, Zare Ghaleh Seyyedi, Abbas, 2023. Power quality enhancement of distribution grid using a photovoltaic based hybrid active power filter with three level converter. Energy Rep. 9, 5432–5448.

Awad, Hilmy, Bayoumi, Ehab H.E., Soliman, Hisham M., Ibrahim, Amr M., 2023. Invariant-set design of robust switched trackers for bidirectional power converters in hybrid microgrids. Ain Shams Eng. J. 14 (9), 102123.

Chen, Xia, Shi, Mengxuan, Sun, Haishun, Li, Yan, He, Haibo, 2017. Distributed cooperative control and stability analysis of multiple DC electric springs in a DC microgrid. IEEE Trans. Ind. Electron. 65 (7), 5611–5622.

De Matos, José G., e Silva, Felipe S.F., de S. Ribeiro, Luiz A., 2014. Power control in AC isolated microgrids with renewable energy sources and energy storage systems. IEEE Trans. Ind. Electron. 62, no. 6 3490–3498.

dos Santos Neto, Pedro J., Barros, T.árcio A.S., Silveira, Joao P.C., Ruppert Filho, Ernesto, Vasquez, Juan C., Guerrero, Josep M., 2020. Power management techniques for grid-connected DC microgrids: a comparative evaluation. Appl. Energy 269, 115057.

Eajal, A.A., Abdelwahed, Mohamed A., El-Saadany, E.F., Ponnambalam, Kumaraswamy, 2016. A unified approach to the power flow analysis of AC/DC hybrid microgrids. IEEE Trans. Sustain. Energy 7 (3), 1145–1158.

Eghtedarpour, Navid, Farjah, Ebrahim, 2014. Power control and management in a hybrid AC/DC microgrid. IEEE Trans. smart grid 5 (3), 1494–1505.

Eisapour-Moarref, Amir, Kalantar, Mohsen, Esmaili, Masoud, 2019. Power sharing in hybrid microgrids using a harmonic-based multidimensional droop. IEEE Trans. Ind. Inform. 16 (1), 109–119.

Elsayed, Ahmed T., Mohamed, Ahmed A., Mohammed, Osama A., 2015. DC microgrids and distribution systems: an overview. Electr. Power Syst. Res. 119, 407–417.

Engler, A. "Control of inverters in isolated and in grid tied operation with regard to expandability in tutorial: Power Electronics for Regenerative Energy." In *Proc. IEEE Power Electron. Spec. Conf., Aachen, Germany* . 2004.

Gupta, Ajay, Doolla, Suryanarayana, Chatterjee, Kishore, 2017. Hybrid AC–DC microgrid: Systematic evaluation of control strategies. IEEE Trans. Smart Grid 9 (4), 3830–3843.

Han, Yang, Xing Ning, Ping Yang, Xu, Lin, 2019. Review of power sharing, voltage restoration and stabilization techniques in hierarchical controlled DC microgrids. IEEE Access 7, 149202–149223.

He, Li, Li, Yong, Guerrero, Josep M., Cao, Yijia, 2019. A comprehensive inertial control strategy for hybrid AC/DC microgrid with distributed generations. IEEE Trans. Smart Grid 11 no. 2, 1737–1747.

Hirsch, Adam, Parag, Yael, Guerrero, Josep, 2018. Microgrids: a review of technologies, key drivers, and outstanding issues. Renew. Sustain. Energy Rev. 90, 402–411.

Hou, Xiaochao, Sun, Yao, Lu, Jinghang, Zhang, Xin, Koh, Leong Hai, Su, Mei, Guerrero, Josep M., 2018. Distributed hierarchical control of AC microgrid operating in grid-connected, islanded and their transition modes. Ieee Access 6, 77388–77401.

Jin, Chi, Peng, Wang, Jianfang, Xiao, Yi, Tang, Fook, Hoong Choo, 2013. Implementation of hierarchical control in DC microgrids. IEEE Trans. Ind. Electron. 61 (8), 4032–4042.

Li, Zhongwen, Cheng, Zhiping, Xu, Yali, Wang, Yifan, Liang, Jing, Gao, Jinfeng, 2019. Hierarchical control of parallel voltage source inverters in AC microgrids. J. Eng. 2019 (16), 1149–1152.

Li, Xialin, Li Guo, Shaohui Zhang, Chengshan Wang, Yun Wei Li, Anwei Chen, Yibin Feng, 2017. Observer-based DC voltage droop and current feed-forward control of a DC microgrid. IEEE Trans. Smart Grid 9 no. 5, 5207–5216.

Liu, Ziwen, Shihong, Miao, Zhihua, Fan, Junyao, Liu, Qingyu, Tu, 2019. Improved power flow control strategy of the hybrid AC/DC microgrid based on VSM. IET Gener., Transm. Distrib. 13 (1), 81–91.

Loh, Poh Chiang, Ding, Li, Yi, Kang Chai, Frede, Blaabjerg, 2012. Autonomous operation of hybrid microgrid with AC and DC subgrids. IEEE Trans. Power Electron. 28 (5), 2214–2223.

Mahmud, Khizir, Sahoo, Animesh K., Ravishankar, Jayashri, Dong, Zhao Yang, 2019. Coordinated multilayer control for energy management of grid-connected AC microgrids. IEEE Trans. Ind. Appl. 55 (6), 7071–7081.

Mohamed, Sayed, Mokhtar, Mohamed, Marei, Mostafa I., 2022. An adaptive control of remote hybrid microgrid based on the CMPN algorithm. Electr. Power Syst. Res. 213, 108793.

Navarro-Rodriguez, Angel, Garcia, Pablo, Georgious, Ramy, García, Jorge, 2018. Adaptive active power sharing techniques for DC and AC voltage control in a hybrid DC/AC microgrid. IEEE Trans. Ind. Appl. 55 (2), 1106–1116.

Nejabatkhah, Farzam, Yun, Wei Li, 2014. Overview of power management strategies of hybrid AC/DC microgrid. IEEE Trans. Power Electron. 30 (12), 7072–7089.

Neto, dos Santos, José, Pedro, Barros, T.árcio André dos Santos, Silveira, Joao Pedro Carvalho, Filho, Ernesto Ruppert, Vasquez, Juan Carlos, Guerrero, Josep M., 2020. Power management strategy based on virtual inertia for DC microgrids. IEEE Trans. Power Electron. 35 no. 11, 12472–12485.

Nordman, Bruce, Christensen, Ken, 2016. DC local power distribution: technology, deployment, and pathways to success. IEEE Electrification Mag. 4 (2), 29–36.

Ordono, Ander, Eneko, Unamuno, Jon, Andoni Barrena, Julen, Paniagua, 2019. Interlinking converters and their contribution to primary regulation: a review. Int. J. Electr. Power Energy Syst. 111, 44–57.

Parhizi, Sina, Lotfi, Hossein, Khodaei, Amin, Bahramirad, Shay, 2015. State of the art in research on microgrids: a review. IEEE Access 3, 890–925.

Phan, Dang-Minh, Lee, Hong-Hee, 2018. Interlinking converter to improve power quality in hybrid AC–DC microgrids with nonlinear loads. IEEE J. Emerg. Sel. Top. Power Electron. 7 (3), 1959–1968.

Planas, Estefanía, Andreu, Jon, Gárate, José Ignacio, Martínez De Alegría, I.ñigo, Ibarra, Edorta, 2015. AC and DC technology in microgrids: a review. Renew. Sustain. Energy Rev. 43, 726–749.

Premadasa, P.N.D., Silva, C.M.M.R.S., Chandima, D.P., Karunadasa, J.P., 2023. A multi-objective optimization model for sizing an off-grid hybrid energy microgrid with optimal dispatching of a diesel generator. J. Energy Storage 68, 107621.

Rahgozar, Sepehr, Zare Ghaleh Seyyedi, Abbas, Siano, Pierluigi, 2022. A resilience-oriented planning of energy hub by considering demand response program and energy storage systems. J. Energy Storage 52, 104841.

Rashid, Sara Mahmoudi, Abbas Zare-Ghaleh-Seyyedi, Jaber Moosanezhad, Ashraf Ali Khan, 2023. Multi-objective design of the energy storage-based combined heat and power off-grid system to supply of thermal and electricity consumption energies. J. Energy Storage 73, 108675.

Rocabert, Joan, Alvaro, Luna, Frede, Blaabjerg, Pedro, Rodriguez, 2012. Control of power converters in AC microgrids. IEEE Trans. Power Electron. 27 (11), 4734–4749.

Sadeghi, Meisam, Khederzadeh, Mojtaba, 2019. Hybrid multi-DC–AC MG based on multilevel interlinking converter. IET Power Electron. 12 (5), 1187–1194.

Sarangi, Swetalina, Sahu, Binod Kumar, Rout, Pravat Kumar, 2020. Distributed generation hybrid AC/DC microgrid protection: a critical review on issues, strategies, and future directions. Int. J. Energy Res. 44 (5), 3347–3364.

Shaban, M., Mosa, Magdi A., Ali, A.A., Abdel-Latif, K.M., 2023. Effect of power sharing control techniques of hybrid energy storage system during fault conditions in DC microgrid. J. Energy Storage 72, 108249.

Silveira, João Pedro Carvalho, Pedro, José dos Santos Neto, Tárcio, Andre dos Santos Barros, Ernesto, Ruppert Filho, 2021. Power management of energy storage system with modified interlinking converters topology in hybrid AC/DC microgrid. Int. J. Electr. Power Energy Syst. 130, 106880.

Tian, Hao, Li, Yun Wei, Wang, Peng, 2017. Hybrid AC/DC system harmonics control through grid interfacing converters with low switching frequency. IEEE Trans. Ind. Electron. 65 (3), 2256–2267.

Tricarico, Thiago, Gontijo, Gustavo F., Aredes, Mauricio, Dias, Robson, Guerrero, Josep M., 2020. New hybrid-microgrid topology using a bidirectional interleaved converter as a robust power interface operating in grid-connected and islanded modes. IET Renew. Power Gener. 14 (1), 134–144.

Ullah, Zia, Wang, Shaorong, Lai, Jinmu, Azam, Muhammad, Badshah, Fazal, Wu, Guoan, Elkadeem, Mohamed R., 2023. Implementation of various control methods for the efficient energy management in hybrid microgrid system. Ain Shams Eng. J. 14 (5), 101961.

Unamuno, Eneko, Barrena, Jon Andoni, 2015a. Hybrid ac/dc microgrids—Part I: review and classification of topologies. Renew. Sustain. Energy Rev. 52, 1251–1259.

Unamuno, Eneko, Barrena, Jon Andoni, 2015b. Hybrid ac/dc microgrids—Part II: Review and classification of control strategies. Renew. Sustain. Energy Rev. 52, 1123–1134.

Wang, Junjun, Jin, Chi, Wang, Peng, 2017. A uniform control strategy for the interlinking converter in hierarchical controlled hybrid AC/DC microgrids. IEEE Trans. Ind. Electron. 65 (8), 6188–6197.

Wen, Sufang, Wang, Shengtie, Liu, Guangchen, Liu, Ruiming, 2018. Energy management and coordinated control strategy of PV/HESS AC microgrid during islanded operation. IEEE Access 7, 4432–4441.

Xia, Yanghong, Wei Wei, Miao Yu, Xiaoming Wang, Power, Yonggang Peng, 2017. management for a hybrid AC/DC microgrid with multiple subgrids. IEEE Trans. Power Electron. 33 no. 4, 3520–3533.
Zong, Shumin, Cao, Juerui, Wang, Can, 2023a. Adaptive normalized droop control for low-voltage hybrid microgrid interlinking converter. Energy Rep. 9, 1377–1388.
Zong, Shumin, Lyu, Yihang, Wang, Can, 2023b. Control strategy of multiple interlinking converters for low-voltage hybrid microgrid based on adaptive droop. Energy Rep. 9, 721–731.